

The Last Human Gate

Forward Deployed Engineering and the
Automation of Enterprise Governance

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Abstract

Can an AI review an enterprise project, identify the problems, and make a justified decision at every governance checkpoint? We introduce DGF-Bench and connect this bounded experiment to a broader thesis about the automation of Digital Governance Frameworks (DGF). The generator constructs 300 fictional projects, their architectures, documents, and simulated records: 100 Buy, 100 Integrate, and 100 Build cases. Agents inspect allowed evidence, apply explicit policies, check mandates, submit structured decisions, and pass their reviews downstream. A separate evaluator scores decisions, findings, proposed actions, evidence, and authorization against the case reference. The tested models are Gemini 3.8 Flash, GPT-5.6 Luna, and DeepSeek v4.1 Flash, accessed through OpenRouter under common task and output limits.

There are 899 evaluable model-case runs out of 900; one provider failure is excluded. Strict gate success is 94.98% for Gemini, 83.29% for Luna, and 74.18% for DeepSeek. Requiring every gate of a project to succeed reduces those rates to 76.92%, 42.33%, and 24.67%. All 85 Gemini strict failures concern evidence conformity, despite correct dispositions throughout its evaluable runs. A deterministic control succeeds on all 1,700 gates under the same supplied rules and facts. An all-model, post-hoc structural evidence audit yields gate success of 99.06%, 85.53%, and 77.24%, without replacing the strict scores. Another 135 runs on 15 existing dossiers test repeated execution: nine Gemini, three Luna, and zero DeepSeek dossiers pass all three strict trajectories. Original and repeated inference cost USD 99.58. These results demonstrate model capability to perform specified review tasks, while establishing no incremental model advantage over the rule program in this information condition. Operational transfer, autonomous remediation, human equivalence, and labor savings are not measured.

The theoretical contribution treats gates as information-transforming contracts and identifies the implementation role of Forward Deployed Engineers. A complete labor account explains why, in a separate synthetic 140-FTE example, identical automation coverage can leave 44 or 169 FTE. The hypothesis of 80% fewer required DGF FTE by 2033, including support at comparable output and quality, remains untested by the benchmark. Complete substitution is a stronger, undated conjecture. Model measurements, workforce scenarios, and prospective claims are kept separate.

Keywords: forward deployed engineering; agentic AI; digital governance framework; governance gates; complete automation; workforce reduction; coordination costs; business process management.

1 Introduction: governance contracts, automation, and human work

An enterprise does not need ten buyers, ten contract lawyers, thirty architects, twenty security architects, and twenty-five production engineers because those numbers are part of its governance. It needs a particular volume of supplier terms negotiated, clauses checked, architectures examined, risks assessed, recoveries tested, budgets validated, and decisions recorded. Those people are the present implementation of that function. An enterprise may still require a contract review, an architecture review, a security review, or a readiness check after it no longer requires a lawyer, an architect, or an engineer to perform each one. The institutional function and its human implementation are different objects.

The relevant replacement is not a model writing a convincing report. It is a system receiving a dossier, obtaining evidence, investigating exceptions, reaching a defensible decision, and making that decision effective under a valid mandate. Drafting assistance is not that replacement. Executing the whole work is, and the population historically needed to run that gate then contracts.

The central conjecture of this paper, termed the law of gate substitution, is that every gate of a Digital Governance Framework (DGF) becomes software, and that the workforce executing those gates is reduced accordingly. Exceptions, verification, arbitration, and final commitment are included. A job title is not a permanent barrier to substitution. Neither is describing a decision as judgment: the question is whether its information, actions, and accepted outcomes can be operationalized. Whatever the role, the input is a set of documents an agent can read and the output is an artifact an agent can write (Section 5). We add, as a separate comparative hypothesis, that the DGF will be the first expert control function to be completely executed by agents (Section 6); the law does not depend on that ranking.

The thesis is also what the author observes in practice while operating such frameworks: some automation programs for governance are already planned and justified in terms of fewer posts and fewer full-time equivalents (FTE), not only faster reviews. The paper takes that objective seriously and asks what it requires.

The path to that endpoint is not a straight line. A company can automate most of its routine reviews and end up needing *more* human work: exceptions take longer, outputs require verification, and maintaining the agents adds an operating burden. Another route on the same platform can save substantial effort. In the synthetic example of this paper, nine specialist populations supply 140 workload-equivalent FTE to a DGF. At the same case volumes, the same exception fractions, and the same selection of cases, alternative operating assumptions leave 43.52 or 168.92 FTE of human workload. Coverage does not identify even the *direction* of the workload change during the transition. The instrument that explains the gap fits on one line:

$$\rho = \frac{\phi\eta + (1 - \phi)\mu + r + b_A}{1 + b_H},$$

the ratio of human work after deployment to human work before it. The share of baseline labor held by the exceptions (ϕ) is multiplied by what deployment does to their effort (η); the rest is multiplied by the review still required (μ); rework (r) and upkeep (b_A) are added (Section 10). A reader who rejects the prediction of this paper can still use this account to evaluate a project. Section 11.3 shows the same fact for two routes of one platform.

The paper therefore does three things. It *argues* for the destination from five premises: contracts already delimit the work; measured agent capabilities are extending to longer tasks; bounded gates already execute as software; authority can be delegated through standing rules; and common platforms can amortize the cost of reliable execution. It supplies the *instrument* that

measures the path: a complete human-labor account, a selection identity explaining why a small exception queue can hold most of the work, a break-even frontier, and fragility thresholds. And it gives the workforce prediction a *deadline*: 80% fewer workload-equivalent FTE required by the DGF ecosystem by 20 September 2033 than in 2026, with failure criteria that cannot be escaped by renaming remaining workers supervisors. Complete automation is the longer-term conjecture, without a fixed date. Workforce displacement follows from successful substitution; it is the consequence of the thesis, not the evidence for it.

The accompanying experiment supplies a narrower, observable contribution (Section 18). We generate a business scenario and its review dossier, ask three models to investigate it through tools, and compare their structured gate decisions with a reference. The same 300 projects yield 899 evaluable model-case runs. This separates two questions that a plausible narrative can obscure: does the agent choose the right decision, and does it provide all the findings, actions, evidence, and authorization required to justify that decision? The experimental section describes the construction, exact model identifiers, tool workflow, scoring, observed examples, results, and limits before drawing conclusions.

The follow-up evidence now includes a deterministic rules control, an exhaustive audit of the 690 gates with failed evidence components, and 135 additional model-case runs on 15 existing dossiers. The control solves the supplied task without a language model; the repeated runs measure within-sample reliability. These findings strengthen the measurement of task execution without making the universal substitution conjecture or the workforce projection an observed result. The same experiment is analyzed in greater empirical detail in the companion manuscript *DGF-Bench: Rule Application and Evidence Reliability in Synthetic Governance Reviews*. The two manuscripts share data and several descriptions of the benchmark; they are not independent replications. This paper’s distinct contribution is the gate-contract argument, formal labor account, organizational mechanisms, and explicitly conditional workforce scenarios.

Scope and status. This paper combines a substitution thesis, a labor-accounting model, and a measured synthetic benchmark (Section 18). The benchmark evaluates a bounded decision-review capability; it does not establish the substitution thesis. No autonomous DGF deployment or enterprise staffing survey is reported. All staffing numbers and calendar scenarios are synthetic calculations. Product documentation establishes described functionality, not its prevalence. Legal provisions constrain the argument and require deployment-specific review. Workload-equivalent FTE measures required effort for a fixed governed output; actual payroll also depends on staffing adjustment, demand, and new tasks. Gates are interchangeable: each enterprise composes its own DGF, and the gates, routes, and orders used here are an arbitrary selection taken as examples. The universal endpoint is a conjecture; the separate 80% FTE-reduction hypothesis for 2033 is deliberately refutable.

2 What a Digital Governance Framework is

A Digital Governance Framework is the set of decision milestones, or gates, that a significant change to an enterprise information system passes before it is allowed to proceed. Whatever its domain, every gate obeys the same mechanics: a gate dossier is prepared by the project team, reviewed in advance by experts, and presented to a validation body that renders a traced decision. Specialized gates issue opinions; a general gate consolidates them and arbitrates the commitment of resources. Table 1 lists eight gate types commonly found in large organizations, with examples of the tasks performed at each.

Four levels of involvement recur at every gate: *deciders*, who hold a deliberative vote and answer for the decision; *dossier owners*, who prepare, present, and defend the dossier; *reviewing experts*, who analyze it beforehand and issue a formal opinion, favorable, favorable with reservations, or unfavorable; and *consulted populations*, mobilized according to the nature and criticality of

Table 1: Eight gate types, what each decides, and examples of its tasks. The list is an example of a gate set, not a standard: enterprises name, merge, split, and order their gates differently.

Gate	What it decides	Examples of tasks	Validation body
General (strategic and budget arbitration)	End-of-phase go or no-go; consolidates the opinions of all other gates.	Check strategic alignment and portfolio priority; update the business case and budget; release funding by tranche; verify that earlier reservations are lifted; record the decision; at closure, validate transfer to run and launch benefits tracking.	Executive or investment committee, sponsor, CFO, CIO, business directors.
IT (technology governance and standards)	Fit with technology standards and the enterprise catalog; operability in run.	Check catalog conformity and process waiver requests, each with a convergence plan; check that no existing application covers the need; review licences, capacity, hosting, obsolescence, and ITSM integration; validate the support model; prepare the change advisory board.	CIO, heads of production and development, IT committee.
Architecture (urbanization and network constraints)	Coherence of the business, application, data, and technical architecture with the information system.	Validate the high-level architecture, then the detailed one; review APIs, flows, interfaces, and data ownership; size bandwidth, latency, addressing, routing, and cloud connectivity; check availability, performance, and reversibility; record structuring decisions and architecture debt.	Chief architect, architecture committee.
Security architecture (cybersecurity and hardening)	Acceptability of cyber risk.	Classify and analyze risk; review segmentation, filtering rules, encryption, identity and access management, hardening, and detection; commission intrusion tests; propose acceptance of residual risks.	CISO, security committee, business risk owner, accreditation authority.
Tech readiness (maturity and technical reliability)	Maturity of the technology and aptitude for production.	Assess technology readiness level and tool stability; review proof-of-concept and pilot results, load tests, and code quality; review the recovery, resilience, and backup dossier; confirm operational aptitude.	CTO, technical director, CIO, production release committee.
Procurement (purchasing, contracts, suppliers)	Sourcing strategy and supplier commitment.	Decide make or buy; run the tender; score offers on multiple criteria and total cost; perform third-party due diligence; prepare supplier contracts and purchasing budgets.	Head of purchasing, commitment committee, CFO, sponsor.
Legal (law and commitments)	Contractual conformity.	Review liability, service-level clauses and penalties, reversibility, intellectual property, data clauses, labor law, insurance, and signing powers.	General counsel, corporate secretary, authorized signatories.
Compliance (standards, regulation, ethics)	Conformity with regulation and standards.	Check GDPR, sector standards such as ISO 27001, PCI-DSS, or health-data hosting, NIS2 and DORA, and the AI Act; review internal control, ethics, accessibility, and archiving.	Chief compliance officer, data protection officer, compliance committee.

the project. The validation body then renders one of five decisions: go; go with reservations, with conditions to be lifted within a set deadline under an action plan; rework, when the dossier is incomplete or insufficient; suspension, pending an arbitration, a budget, or an external prerequisite; and no-go. Risk is scored as ordinal impact multiplied by ordinal likelihood, each from 1 to 4; a score of 12–16 is critical and calls for rejection or competent governance arbitration, and an above-tolerance acceptance requires a business risk owner and a dated acceptance record (Canale, 2026, pp. 38, 98, 100–101). The person who must fix a defect is therefore distinct from the person empowered to accept the business risk. Composition is proportional: a simple

business SaaS project does not mobilize the same experts as a critical infrastructure program, job titles vary, and in mid-size structures one person often holds several roles.

2.1 Gates are interchangeable: each enterprise composes its own DGF

A DGF is composed, not prescribed. Each enterprise decides which gates belong to its DGF, on which routes, in which order, and with which tasks; no gate of Table 1 is mandatory, and none is missing by nature. Gates are interchangeable in a precise sense: they are modular and reconfigurable. Because every gate has the same form, a dossier prepared, reviewed, and decided, an enterprise can add, remove, merge, split, or move a gate. It cannot do so freely. A recomposition must preserve the obligations, the informational prerequisites of each gate, and the authorization relations between them: a gate that assesses a scope cannot be moved before the gate that defines that scope without changing its contract, accepting a provisional result, or providing a return. A common form does not make two contracts identical, and Section 13 shows how much a gate depends on what arrives from upstream. Three things therefore vary from one enterprise to another, and within one enterprise from one project to another. *The set of gates varies.* Some organizations merge Architecture and Security into one review; others add gates for data, accessibility, or sustainability. *The order varies.* Table 2 shows one project life cycle in which the same gate is mobilized in several phases, with a different task each time: Security classifies and analyzes risk at framing, reviews the security dossier at design, and runs intrusion tests before production; Architecture validates a high-level architecture, then a detailed one. In agile delivery the gates are kept but lightened: controls are spread across increments, and only major commitments give rise to a formal passage. *The tasks vary.* Third-party due diligence sits in Procurement here and in Compliance elsewhere; a licence review belongs to the IT gate in one enterprise and to Procurement or Legal in another.

Table 2: One possible positioning of the gates in a project life cycle. The same gate returns in several phases with different tasks; the general gate closes each phase.

Project phase	Specialized gates mobilized, with their task in that phase	General gate decision
1. Opportunity	IT: does a solution already exist in the catalog? Compliance: regulatory sensitivity of the subject.	Go for study: authorization to frame.
2. Framing and feasibility	Architecture: high-level architecture. Security: classification and initial risk analysis. Tech readiness: maturity of candidate technologies. Procurement: purchasing strategy. Compliance: applicable regulations.	Investment go: business case and budget envelope validated.
3. Design	IT: catalog conformity. Architecture: detailed and network architecture. Security: security dossier. Procurement: supplier choice. Legal: contracts. Compliance: impact analysis.	Go for build: construction spending committed.
4. Build and acceptance	Tech readiness: tests, quality, operational aptitude. Security: intrusion tests. IT: operations dossier, change advisory board. Compliance: evidence of conformity.	Go for production: deployment authorized.
5. Deployment and closure	IT: transfer to run. Procurement and Legal: acceptance, warranties, reversibility. Compliance: recurring controls.	Project review: closure, lessons learned, benefits tracking.

The gates, routes, and orders used in this paper are therefore an arbitrary selection, taken as examples. The three routes of Section 3, Buy, Integrate, and Build, each mobilize a different subset of the eight gate types in a different order; another enterprise would draw them differently. Nothing in the argument depends on that selection. The law of gate substitution applies to a gate because of its mechanics, a dossier read and a traced decision returned, not because of its name, its position, or its task list; it applies equally to gates that do not appear in this paper. Law 3 asks automation to preserve the composition the enterprise has decided, not a

universal sequence: the enterprise stays free to recompose its DGF, and the automation does not recompose it to make its own task easier.

A DGF is a *paved road*: a standardized route with known artifacts, decision rules, interfaces, and escalation paths. Standardization is what makes it governable by humans today, and it is also what makes it executable by software tomorrow: instead of reconstructing an open-ended organization, an agent receives a bounded case, a policy version, tools, and an output contract.

The DGF is staffed by specialist populations, not by one department. Table 3 gives the illustrative workforce used throughout the paper. A population can contribute to several gates, and a gate can require several populations. R1–R9 are resource identifiers, not nine sequential approvals. Let P_d denote people in department d , K_d their useful hours per period, and α_{di} their allocation to work package i . Allocated capacity is $\sum_d P_d \alpha_{di} K_d$, with $\sum_i \alpha_{di} \leq 1$ across all routes and shared work. Thirty architects spending half their useful time on these activities supply fifteen allocated FTE. Approval and shared-assurance hours are counted once; the sponsors’ commitment at the general gate is a distinct authority rather than an extra finance population.

Table 3: One illustrative workforce, aligned with the gate types of Table 1 and shared across three generic routes. The contributions are an analytical mapping; the headcounts are synthetic.

ID	Population (gate)	People	Buy (W1)	Integrate (W2)	Build (W3)
R1	Procurement	10	Sourcing strategy, tender, supplier due diligence.	Inherited supplier contracts and provider evidence.	External components and services, if any.
R2	Legal / Contracts	10	Contract, data-processing and service-level clauses.	Inherited obligations, novation, reversibility.	Licences, intellectual property, data-use terms.
R3	Architecture	30	Consulted on integration and data flows; no Architecture gate on this route.	Mapping of the inherited system, target integration.	High-level and detailed architecture.
R4	Security Architecture	20	Supplier security package, exposure, controls.	Acquisition audit, exposure, risk conditions.	Design assessment and security conditions.
R5	Compliance / Privacy / GRC	10	Processing assessment, transfers, sector rules.	Inherited data and regulatory obligations.	Impact assessment and applicable rules.
R6	IT Production and Standards	25	Catalog fit, run and support model.	Inventory, rationalization, obsolescence, run takeover.	Catalog conformity, operations dossier, change board.
R7	Tech Readiness	15	Pilot results, maturity of the service.	Stability of the inherited platforms.	Tests, load, code quality, production aptitude.
R8	General Gate Support (PMO, portfolio, risk)	10	Gate dossier, reservations register, schedule, and risk cards on every route.		
R9	Finance / Cost Control	10	Pricing, subscription cost controls.	Integration-option costs.	Project budget and cost assumptions.
Total		140	One shared pool, not 140 people for each route.		

No population is the principal actor of a DGF. Buyers and lawyers open a purchase; production and standards teams open an integration or a build by asking what already exists; architects, the largest population of the illustrative pool, shape every route; security architects issue one opinion among several; compliance officers, readiness experts, portfolio, risk, and finance staff assemble the decision that the general gate takes and that sponsors commit to. The law of gate substitution applies to each of these roles in the same way, and the examples below deliberately

draw on different populations (Section 5).

3 Three example routes built from interchangeable gates

Three generic routes serve as examples throughout the paper. **Buy** acquires a platform or service from a supplier. **Integrate** absorbs an information system that already exists, for example after an acquisition. **Build** develops a new internal solution. They abstract the three route types of the author’s practitioner framework, a new platform, a merger or acquisition, and a new project (Canale, 2026, pp. 17–22). Figure 1 draws one possible composition for each. They are three compositions among many: another enterprise could drop Compliance from Integrate, add Tech readiness to Buy, put Security architecture before IT in Build, or use gates that are not in Table 1 at all.

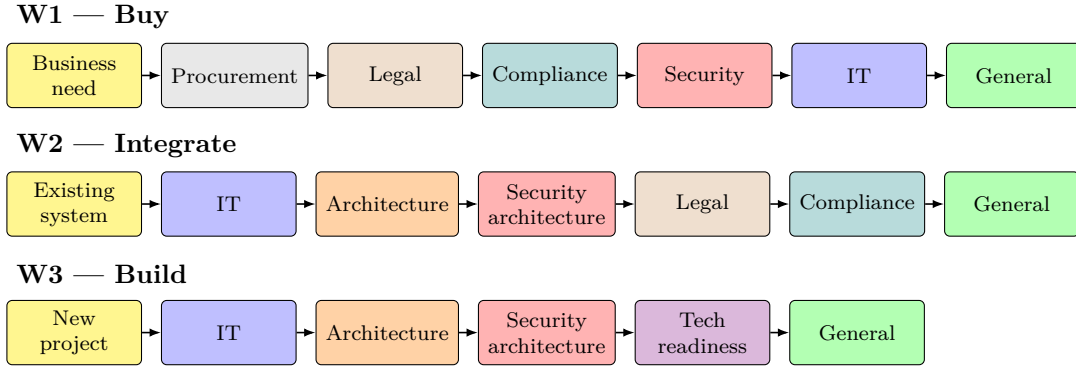


Figure 1: Three generic DGF routes, one illustrative configuration. Yellow: initiating need or event; grey: Procurement; brown: Legal; teal: Compliance; red: Security, a supplier security review in Buy and Security architecture elsewhere; orange: Architecture; blue: IT; purple: Tech readiness; green: General gate. Colors identify gate types, not automation maturity. Each route mobilizes a different subset of the eight gate types of Table 1, in a different order, and ends at the general gate. Enterprises define their own gates, orders, and tasks.

The triggering need, system, or project is not another approval gate. In Buy, the supplier’s own contribution, its proposal, evidence, and commitments, enters through Procurement; it belongs to another organization and cannot be removed from an end-to-end automation analysis. In this configuration the three routes contain seventeen gate occurrences after the trigger: six in Buy, six in Integrate, five in Build. Every gate type appears at least once. The sponsors’ commitment is recorded at the general gate; it is not drawn as a separate box.

The order differs for a reason in each example. Buy examines the supplier’s commitments, contract, data obligations, and security before the run model is fixed; its security gate is a supplier security review, and no dedicated Architecture gate is mobilized: architects are consulted on integration and data flows without a gate of their own. Integrate starts from what exists: IT inventories the inherited system before Architecture and Security assess it, and Legal and Compliance then qualify the inherited obligations. Build starts by asking whether a solution already exists in the catalog, and adds a Tech readiness review before production. These orders are examples (Section 2.1): the enterprise chooses its gates and their order, and what the law requires is that automation preserve that choice, not these particular routes.

The figure shows nominal forward paths. Reservations, rework, suspension, evidence requests, and dated conditions remain part of execution, and several gates can review one shared dossier in parallel. An implementation that always approves, or always refuses to avoid hard cases, fails the function. Full automation preserves substantive decision quality and the permitted control flow, not just the order of labels. The same label can also describe different context-sensitive

transformations: the security gate in Buy evaluates a supplier’s package and commitments; in Integrate, Security architecture evaluates an inherited system already scoped by IT.

4 A gate is a contract, not a profession

Definition 1 (Gate-centered work package). A work package $G_{w,j}$ is the preparation, inquiry, evaluation, decision production, authorized action, and handoff surrounding position j of route w . For versioned state S , accessible evidence E , and policies Π ,

$$(S^+, D, U) = G_{w,j}(S, E; \Pi, w), \quad (1)$$

where D is a disposition and U an action/evidence trace. The procedure may be interactive and stochastic.

The input includes more than a PDF: architecture records, observations obtained through tools, a supplier’s reply, costs, identities, outstanding conditions, and a history of negotiations. Requesting missing evidence is part of the transformation. The output includes more than prose: it can create a ticket, block a connection, require a redesign, approve a bounded action, or record a refusal. A valid negative decision is an output; an unsupported declaration that an absent report was verified is not. Authorization makes an eligible disposition effective; producing it does not confer that authority.

A contract $C_{w,j}$ specifies admissible input histories, acceptable output and action relations, evidence obligations, quality tolerances, and authority. Several decisions may be defensible. A replacement need not reproduce a particular person’s wording, but it must meet the accepted decision standard. The reference standard also records disputed judgments instead of treating every historic human decision as ground truth.

Definition 2 (Replacement of gate execution). An implementation replaces a gate when it satisfies $C_{w,j}$ without requiring a person to complete its ordinary work, exceptions, verification, or per-case authorization. Human participation that remains necessary is counted as residual execution labor, whether performed internally, by a contractor, or behind an apparently autonomous service.

This distinction is established organizational practice, not a newly discovered property of computation. Stage-Gate separates the work performed in stages from gates that evaluate deliverables, apply criteria, and decide whether a project proceeds, waits, stops, or returns for further work (Cooper, 1990, p. 46); artifact-centered process modeling represents the records transformed by business operations (Nigam and Caswell, 2003). These antecedents support implementation neutrality. They do not by themselves prove that every information contract is computable, sufficiently observable, economically executable, or legally delegable; Sections 6 and 7 give the reasons to expect it for the DGF.

4.1 Three conditions for complete execution

For a declared case population and contract version, complete execution has three simultaneous conditions. *Labor*: no necessary human time on any admitted visit, including exceptions and per-case authorization. *Quality*: independently adjudicated performance satisfies prespecified noninferiority margins against the human reference and absolute limits on consequential errors, false refusals, information leakage, and delay. *Mandate*: the decision is institutionally effective, not merely a recommendation.

A sample containing no human interventions supports, but cannot prove, zero intervention on every future input. Operational evidence must therefore combine time records, tests, and an

architecture with no required manual step. Assessment by researchers does not itself make operation manual; any continuing human assessment required to keep the service qualified does count as support. The distinction is functional, not organizational. A service with autonomous visits but a human maintenance team has achieved execution closure, not support closure. Initial development is recorded as a transition investment; continuing updates, incident response, mandated oversight, and outsourced verification cannot disappear from the account because they have moved to a supplier.

4.2 One handoff contract and four operating stages

A supplier contract can reach Legal as a PDF, Security as an email attachment, and IT production as a project-manager summary. Each function then extracts the supplier, hosting locations, security commitments, unresolved clauses, and evidence expiry dates. The interface gain is to perform this extraction once and attach a provenance trail that the other functions can inspect. The minimal reusable record contains case and source version, route and gate, claim, evidence locator, verification status, action owner, risk owner, deadline, and authorization reference:

```
route: W3
current_gate: TECH_READINESS
recommendation: NOT_READY
claim: load_test_incomplete_and_restoration_not_tested
evidence: {reference: restore-test-report, status: missing}
action_owner_role: production_lead
risk_owner_role: business_service_owner
approval_reference: null
execution_authorized: false
next_action: run_restoration_test_or_request_authorized_exception
```

The record is a schema illustration. Its empty approval reference blocks authorization. A field marked missing carries information; replacing it with an invented value destroys that information. This is the difference between a local copilot and a connected process: a copilot saves one reviewer's reading time, whereas a dependable evidence object changes the work of several downstream functions. A human can also create that object, which is why Section 17 separates the value of structure from the value of agentic production.

In a **human pipeline**, specialists prepare and check every dossier. In an **assisted pipeline**, agents collect or draft while humans inspect the ordinary path. In an **exception-routed pipeline**, eligible work executes automatically, while experts handle exceptions, sampled review, and retained authorizations. The thesis adds a fourth stage, the **closed pipeline**, in which exceptions, verification, and authorization are executed under mandate as well. The configurations can coexist within a route, and no company is assumed to move monotonically: incidents can restore manual review. The transition moves the expert from reviewing every ordinary case to governing the system that reviews them (Section 19), until those tasks are themselves executed by software.

5 Every role has the same shape: documents in, an artifact out

Whatever the role, the work of a DGF has one shape. The input is a set of documents and records that an agent can read: a design, a diagram, a contract, a test report, a register of findings. The output is an artifact that an agent can write: an opinion, a list of findings, a redlined clause, a readiness verdict, a risk card, a decision record. A security architect reviewing an architecture diagram, a readiness expert reviewing a recovery, resilience, and backup dossier, and a GRC expert proposing a risk card perform the same kind of transformation. What differs

between them is the policy applied and the evidence required, not the form of the work. Table 4 states that shape for every population and for the general gate.

Table 4: Documents in, artifact out: the input each role reads and the output it returns.

Role	Input the agent can read	Output the agent can return
Procurement (R1)	Business need, vendor proposals, price lists, supplier questionnaire.	Supplier comparison, normalized commercial terms, sourcing recommendation, contract draft.
Legal / Contracts (R2)	Contract draft, data-processing agreement, SLA and security annexes, subprocessor list.	Redlined clauses, accepted and rejected terms, legal exceptions, obligations register.
Architecture (R3)	Requirements, high-level and detailed architecture, integration and flow diagrams, network matrix.	Architecture opinion, integration pattern, network conditions, recorded decisions and waivers.
Security Architecture (R4)	Architecture diagram, flow matrix, data classification, identity and access model, supplier evidence.	Findings with severity, required controls, risk scenarios, opinion.
Compliance / Privacy / GRC (R5)	Data categories, purposes, processors and locations, control assessments, gate findings.	Assessment requirements, transfer and retention conditions, risk card with owner and treatment.
IT Production and Standards (R6)	Catalog conformity sheet, licence inventory, operations dossier, support model.	Conformity opinion, waivers with a convergence plan, run acceptance.
Tech Readiness (R7)	Pilot, load-test, and code-quality results; recovery, resilience, and backup dossier: backup policy, restoration test report, recovery objectives, runbooks.	Readiness verdict, open conditions, retest dates.
General Gate Support (R8)	Gate opinions, reservations register, schedule, risk cards, residual risks.	Consolidated gate dossier, dated action plan, acceptance package.
Finance / Cost Control (R9)	Solution design, usage assumptions, quotes, budget envelope.	Budget opinion, cost guardrails, variance alerts.
General gate (committee and sponsors)	Gate dossier, opinions, residual risks, budget, schedule.	Recorded decision: go, go with reservations, rework, suspension, or no-go, with owners, dates, and the sponsors' commitment.

The three worked routes of Section 9 come from a security practitioner's framework, so their blocking findings are security findings. Table 5 therefore adds eight examples, one per gate type of Table 1, in which the blocking finding arises at a different gate each time. They are illustrative examples written for this paper, not cases from the framework.

Three remarks follow. First, when a document is missing or insufficient, the output is a request for evidence, an unresolved status, or a refusal; that is still an artifact, and producing it is part of the contract of Definition 1. Second, some inputs are obtained rather than received: a restoration test is launched, a supplier is asked, a configuration is queried. Those are actions an agent can take through tools, within its mandate. Third, the artifact of one role is the document of the next: the IT opinion is read by Architecture and Security, the findings are read by Compliance, the risk card is read by the committee, the decision is read by the sponsors. A DGF is a chain of documents read and artifacts returned, which is the form of work that agents already perform. This is the practical content of Law 1.

The shape of the work is not a proof that all its requirements can be executed. Two situations

Table 5: Eight illustrative reviews, one per gate type. In each, the expert reads documents and returns an artifact; nothing in the transformation requires that the reader be a person.

Gate and role	Documents read	Blocking finding	Artifact returned
General: portfolio committee	Gate dossier: opinions of the specialized gates, risk cards, updated business case, budget, schedule.	Cost above the approved envelope and one reservation still open, while the business asks to proceed.	Recorded decision: go with reservations under a budget cap, a named risk owner, and a review date, or no-go.
IT: production and standards lead	Catalog conformity sheet, licence inventory, operations dossier, support model.	Database engine outside the catalog and out of vendor support; no on-call model defined.	Opinion with reservations: waiver request with a convergence plan; support model required before the change advisory board.
Architecture: enterprise architect	High-level architecture, flow diagram, interface list, network matrix.	Point-to-point file transfer bypassing the integration platform; inter-site bandwidth insufficient for nightly volumes.	Architecture decision record: required integration pattern, network sizing condition, waiver recorded with a remediation date.
Security architecture: security architect	Architecture diagram, flow matrix, data classification.	Administration interface exposed to the internet; unencrypted flow carrying confidential data.	Findings with severity, required controls, unfavorable opinion or opinion with reservations.
Tech readiness: readiness expert	Pilot and load-test results, code-quality report, recovery, resilience, and backup dossier, runbooks.	Load test stopped at half the target volume; last restoration test older than twelve months.	Not-ready verdict, tests requested with dates, handover blocked until the evidence is attached.
Procurement: buyer	Need statement, supplier proposals, pricing grids, third-party due-diligence report.	Lowest bid has the highest total cost once licences and exit costs are counted; one supplier fails financial due diligence.	Multi-criteria scoring, sourcing recommendation, negotiation points for the commitment committee.
Legal: contract lawyer	Supplier contract, data-processing agreement, service-level annex.	Liability cap below policy; no audit right; subprocessor change without notice.	Redlined clauses, legal exceptions with positions, obligations register for the supplier file.
Compliance: GRC expert	Gate findings, control assessment, processing description, applicable regulations.	Residual risk above tolerance after the proposed controls; impact assessment missing for a new processing of personal data.	Risk card: scenario, impact and likelihood classes, score, treatment options, proposed risk owner, acceptance record to be signed; impact assessment required.

that look identical in the accessible documents can require different decisions; no reader, human or machine, can then decide correctly without further inquiry (Appendix C). The shape says where inquiry must go and what must be returned; whether every requirement can be met without a person is what the hypotheses of Section 8.3 test.

6 Why the DGF is the most likely first candidate for agentic automation

Alongside the law, we state a separate comparative hypothesis: **among the expert control functions of an enterprise, the DGF will be the first to be completely executed by agents.** The law does not depend on it; the DGF could close after another function and the law would still hold. The hypothesis rests on seven properties that few other functions combine.

Its objects are already digital. A DGF reviews software systems, configurations, contracts,

evidence, and institutional commitments. Repositories, identity systems, cloud control planes, ticketing tools, and APIs already supply most of the observations and actions a gate needs. Nothing has to be lifted, driven, or examined at a bedside: every role reads documents and returns an artifact (Table 4).

It is already a paved road (Section 2). The representation work that blocks automation elsewhere has been done here for human reasons.

Its decisions concern systems and projects. An architecture approval is not a hiring, credit, or medical decision about a person. This does not exempt it from law (Section 7.3), but it places the DGF further from the strictest regimes on automated decisions than most candidate functions.

It is a cost center and a bottleneck. Review queues delay every project of the enterprise. An internal review is not a product feature: customers buy the firm’s products, not the identity of whoever performed its internal reviews. Where a contract or a regulator does require a named human reviewer, that is an institutional constraint of the kind discussed in Section 7.3, not a commercial preference. Management therefore faces both a speed incentive and a cost incentive.

Its work is specialist work, and much of it is repetition. The same facts are extracted from the same documents by Procurement, Legal, Security, and IT production. How many visits are standard is an empirical frequency, measurable on historical logs. Hypothesis H3 makes a separate prediction: the residual holds a larger share of the hours than of the cases (Section 10.2), whether or not standard cases are the majority.

Its outputs are checkable downstream. A condition such as “EDR before interconnection” or “log export in the contract” can be verified by the next gate or by the production system itself. A function whose outputs can be tested is a function whose automated executor can be qualified.

Its neighbors are already software. Bounded governance decisions around the DGF already execute without a fresh human judgment (Table 7).

The comparison class matters. Transactional back-office functions, such as invoice processing, order creation, post-trade matching, and first-level service desks, are already largely executed by software; they are the precedents of Section 7.2, not rivals for this hypothesis. The rivals are the other expert control functions staffed by senior specialists: legal review, financial control and audit, credit and underwriting decisions, human-resources decisions, clinical review. The criterion is execution closure as defined in Section 4.1, reached for every gate of the function at a participating site. Human-resources, credit, and clinical decisions concern persons and meet the strictest legal regimes; legal work carries privilege and professional liability; audit is bound by independence rules. The DGF combines high structure, digital evidence, internal impact, and management pressure. The hypothesis fails if another expert control function reaches execution closure first. On this hypothesis, the populations of Table 3 are the first specialist workforce exposed.

7 Why believe complete automation will extend to every gate?

The argument is inductive and compositional, rather than a deduction from notation. Table 6 distinguishes evidence already available from the extension that the thesis needs.

7.1 Capability is moving toward longer coherent work

Kwa et al. (2025) introduce a task-completion horizon measured in human expert time, not in the agent’s runtime. Their 2019–2025 results show an approximately seven-month doubling of the

Table 6: Five premises, with their evidential boundary. The final column contains the substantive burden of the complete-automation thesis.

Premise	Evidence or mechanism	Remaining claim to test
Contract, not occupation	Each DGF specifies evidence, criteria, outputs, and the order chosen for its routes.	Contracts capture material inquiry and judgment without concealing a necessary human.
Capability reaches gate scale	Measured software-task horizons have increased substantially.	Comparable progress transfers to messy gate work and the required reliability tail.
Gates already become software	CI quality rules, cloud policy, standard changes, and transaction processing execute bounded controls.	Automation extends from these controls to the full judgment and exception loop.
Authority becomes a mandate	Authorized rules can release changes or orders without another per-case decision.	Required commitments can be validly delegated on each DGF route.
Provision scales	Reusable services can spread common engineering and assurance costs.	Local adaptation and risk costs do not preserve an economic or human-support floor.

50% horizon on software-related tasks, and a similar trend for the 80% horizon at roughly one fifth of the length. They associate the increase with better reliability, error recovery, reasoning, and tool use. This is the strongest measured trend supporting the move from individual instructions to extended work packages.

The later METR dashboard, updated on 8 May 2026, reports for its best-performing model a 50% horizon of at least sixteen hours, with a 95% interval of 8.5–55 hours, and warns that measurements above sixteen hours exceed the reliable range of its task suite (METR, 2026a). The reported point is not a guarantee that every sixteen-hour activity can be completed.

The inference to gates requires care. A twelve-, twenty-four-, or thirty-six-hour dossier can contain parallel work by several specialists; it is not automatically one coherent task of that duration. The 12–36-hour examples below are synthetic reference workloads, not measured DGF horizons. METR’s tasks are mainly self-contained software engineering, machine-learning, and cybersecurity problems; its documentation emphasizes domain variation over orders of magnitude, low-context human baselines, worse transfer to messy work, and error bars of roughly a factor of two (METR, 2026a,b). It does not estimate 99% horizons.

These qualifications change how the evidence is used, not its relevance. A trend from short tasks to multi-hour, tool-using work is evidence against a fixed boundary at a document summary. The thesis is that the same combination of improved reasoning, access, recovery, and verification closes the larger contracts. Sections 15 and 17 expose the additional assumptions and tests.

7.2 The inductive precedent: some decisions already run as software

Table 7 documents operational mechanisms in the same organizational neighborhood as the DGF. The important observation is not a vendor’s adoption claim. It is that an input can already be checked against an authorized condition and cause an effective action without a fresh human judgment in that bounded case.

These precedents refute the proposition that a governance-related decision necessarily requires an employee to handle every instance. Some implement the decision itself; others automate evidence or recommend an outcome while retaining a human. That contrast is useful: it identifies which part of the contract remains open instead of labelling every use of software a completed gate.

Table 7: Existing automation near the gate boundary. Documentation supports the stated mechanism, without implying zero supporting labor or full automation of the wider function.

Primary example	Software-executed transformation	Boundary of the precedent
CI/CD quality gates (SonarSource, 2026)	Code-analysis measurements and conditions produce pass/fail; pipeline or merge policy can enforce the result.	Encoded quality criteria, not every design or security judgment.
Cloud policy-as-code (Microsoft, 2025)	A resource request matching a deny policy is rejected before resource-provider execution.	Enforces a rule; does not establish that the rule covers all risks.
Standard changes (Atlassian, 2026)	A repeatable, documented, low-risk change can be pre-authorized rather than approved anew.	Preauthorization supplies the mandate; automated execution still needs an implementation.
Continuous deployment (Amazon Web Services, 2026a)	A qualifying change proceeds automatically into production, unlike delivery awaiting explicit release approval.	Tests, configuration, rollback, and supporting operations remain relevant.
Touchless purchasing (Oracle, 2026)	An approved requisition can become a purchase order and be communicated without a procurement agent’s intervention.	Requisition approval and agreement formation are prior inputs, not shown to be autonomous.
Compliance evidence (Amazon Web Services, 2026b)	Service data are collected and associated with control evidence.	The service explicitly does not assess compliance itself.
Post-trade processing (DTCC, 2026)	Matching enables automatic trade affirmation and downstream processing.	A bounded post-trade path, not the entire back office or all exceptions.
Automated underwriting (Fannie Mae, 2026)	Application and credit information produce a risk assessment and recommendation.	The lender retains underwriting judgment and responsibility.

The next step is therefore an extension, not a new direction of history. Deterministic controls already translate accepted requirements into executable tests. Agents add interpretation, evidence pursuit, exception diagnosis, and revision of a proposed plan. The favorable induction is that a growing composition of such operations replaces the whole gate. Its strongest rival is that unstructured judgment or assurance remains an irreducible last task. Section 16 tests that rival rather than assuming it away.

7.3 From an individual signature to a standing mandate

Authorization is not the same as a handwriting event. Continuous deployment and touchless order creation illustrate execution under rules and prior approvals (Amazon Web Services, 2026a; Oracle, 2026). A DGF can similarly move an eligible class of decisions from repeated individual approval to a standing delegation. The delegation specifies case class, allowed actions, budget and risk limits, evidence requirements, expiry, suspension, and the principal on whose behalf the system acts.

At the general gate, the go decision and the sponsors’ commitment are then a recorded, valid exercise of that mandate. The gate remains and the commitment is effective; no person is impersonated and no machine declares itself a corporate officer. Cases outside the mandate remain unresolved or are refused under the contract. A mandate with a required human exception path has only partially automated the gate; its extension to every case class is part of the thesis.

The legal limit is substantive. NIS2 Article 20 requires management bodies to approve and oversee cybersecurity risk-management measures and addresses their liability and training.

DORA Article 5 assigns management bodies responsibility for the ICT risk-management framework; Article 6(10) permits outsourcing verification tasks while retaining the financial entity’s responsibility (European Parliament and Council, 2022a,b). These texts support distinguishing execution from responsibility, but do not authorize eliminating all human governance. Required continuing board work belongs in B_A . Under a regime that retains it, a zero-support claim for that broad boundary fails unless the requirement itself validly changes.

Nor is “a decision about systems” a general exemption. GDPR Article 22 concerns solely automated decisions about people with legal or similarly significant effects, subject to specified exceptions and safeguards; ordinary architecture approval is not automatically such a decision, but a DGF service can affect access, employment, or other protected interests (European Parliament and Council, 2016). The AI Act’s high-risk classification and human-oversight provisions also depend on intended use and applicable conditions, not the label DGF (European Parliament and Council, 2024, Arts. 6, 14 and Annex III). Jurisdiction, national implementation, effective dates, and sector rules must be assessed for the actual service. No legal opinion or universal permission is asserted here.

The argument for migration is thus operational: the authorized principal can define a class-level rule and let software execute it. The argument against complete support closure is institutional: some continuing human obligations may survive that migration. Both are observable. Maintaining this distinction strengthens the automation thesis by identifying what must change, rather than treating liability as either a magical human advantage or an irrelevant detail.

7.4 Shared provision can make a low-volume route economical

A five-visit Integrate route may be uneconomic when it must build and maintain its own agent platform. That does not imply that it remains uneconomic when buying a common service. Let F_s be fixed, shareable nonlabor platform cost, N comparable customers, F_c customer-specific fixed cost, a variable nonlabor cost per visit, and λ_c visits at customer c . Under an equal allocation convention,

$$\bar{c}_{c,\text{platform}} = a + \frac{F_s/N + F_c}{\lambda_c}. \quad (2)$$

The common term falls with scale, while local integration does not. Falling provider cost reaches the customer only if pricing passes through some of the saving. The same distinction applies to human upkeep. Allocating shared hours B_s and customer-specific hours B_c gives

$$b_{A,c} = \frac{B_s/N + B_c}{\lambda_c h_{0,c}}. \quad (3)$$

Table 8: Synthetic low-volume provision: 60 common human upkeep hours per month plus six customer-specific hours, five visits per customer, and 24 baseline hours per visit. Provider labor has not disappeared; it is allocated explicitly.

Customers	Support hours/month/customer	Support hours/visit	Total hours/visit	ρ
1	66.00	13.200	34.820	1.451
10	12.00	2.400	24.020	1.001
100	6.60	1.320	22.940	0.956
1,000	6.06	1.212	22.832	0.951

Table 8 applies Equation (3) to the five-visit Integrate route of Section 11.3, whose direct residual effort is 21.62 hours per visit. Alone, the customer faces a workload ratio of 1.451; sharing the sixty common hours across 1,000 customers while keeping all six local hours lowers it to 0.951. The verdict on a low-volume route therefore turns on whether it must carry its own policies,

tests, and evaluations or can share them. The table holds the sixty common hours constant as customers are added, which is the favorable case: the empirical question is whether shared support hours grow less than proportionally with the number of customers.

Amortization lowers a buyer’s allocated burden; it does not eliminate the provider’s workers. Across all customers the shared hours still sum to B_s , and the local hours to $\sum_c B_c$. Full support closure requires automating or removing the required work itself. Economies of scale make the investment plausible; they are not a shortcut to zero global labor.

8 The law of gate substitution: statement, five component laws, six path hypotheses

The law of gate substitution. Every gate-centered work package that an enterprise places in its DGF, whatever gates it has chosen, obtains, and moves to, an effective automated implementation of its accepted input-output contract. Inquiry, exceptional-case analysis, verification, the general gate’s arbitration, and the commitment it records are included. The recurring work needed to operate and verify those implementations is itself a substitution target, not a permanently human role. At fixed governed output, the human workforce required to execute the DGF contracts toward its remaining support floor, and then toward zero.

Status: a conjecture proposed by the author, not an established result. The separate dated hypothesis of an 80% FTE reduction by 2033 (Section 17) can be refuted without resolving this longer-term conjecture.

“Every gate” is a universal claim over gate types and their declared case domains. It is not a claim that a machine will know an unobserved fact, that every judgment is error-free, or that all enterprises cross the boundary on the same date. The accepted quality specification stays fixed while implementations are compared; the thesis does not make the specification easier by removing difficult cases. Its technical and adoption components are separately observable: a qualifying implementation exists, and the organization actually uses it under a valid mandate.

8.1 Five component laws

The five laws name distinct commitments of the thesis.

Law 1 — Contract substitutability. A DGF gate’s function is specified by the transformation it performs, not by the profession that currently staffs it. Operational contracts, augmented with inquiry and action interfaces, capture enough of that function to support an implementation-independent evaluation. It fails for a stated domain if a material part of successful execution cannot be represented or evaluated without an indispensable personal intervention.

Law 2 — Complete execution. No DGF gate class is a permanent exception to autonomous execution. Progressively broader implementations ultimately cover the entire gate, rather than stopping at a permanently human layer of difficult decisions. A durable technical or institutional requirement for human execution in one gate class contradicts its strong form.

Law 3 — Route preservation. Replacing the execution of every component replaces the execution of the route without erasing its required gates. Replacements preserve the composition the enterprise has decided: its gates, their order on each route, their conditions, and their returns. Gates are interchangeable for the enterprise that composes the DGF, which can add, remove, merge, or reorder them at any time; they are not interchangeable for the agent

that executes it. What is preserved is every obligation and the logical order of authorizations, not the physical existence of each box: one service can execute several contracts, but dropping or reordering a required control to make automation easier is not its replacement.

Law 4 — Execution displacement. For a fixed amount of governed work, automating the complete route removes the labor demand that staffed its execution. The reduction is not merely a change in job descriptions. It follows when removed execution work is not replaced by equally large required human support (Section 12).

Law 5 — Supervisory closure. Monitoring, exception review, policy maintenance, and verification are additional transformations to automate, not permanent human refuges assumed by definition. If they remain labor-intensive, the system is only partially closed.

8.2 What makes the law stronger than an augmentation claim

An augmentation claim permits the same human team to remain indispensable while using better tools. Complete substitution does not. A staff member who must interpret every output, reconstruct every rare incident, or sign every individual approval is still executing part of the gate. Reducing the average time of that intervention is progress toward substitution, not completion of it. Likewise, the future workload is not assumed to consist only of easy cases. When an exception becomes a named subprocedure, the thesis treats that subprocedure as a new target for the same input-output analysis. An indefinitely expanding collection of indispensable human subprocedures would defeat the strong endpoint, even if each software release automates more standard cases.

8.3 Six hypotheses about the path

The component laws state the destination. Six hypotheses describe the transition and can be tested long before the endpoint. Each predicts an observable pattern and can fail within a specified case population. Automation is evaluated against independently assessed quality, not merely whether a workflow reaches an approval state.

H1 — Gate readiness. Gate-level readiness improves out-of-sample prediction of reliable automation beyond model capability and case length. Accessible inputs, explicit criteria, independent evidence checks, scoped actions, and testable outcomes explain differences between otherwise comparable gate implementations (Appendix B).

Failure criterion: prospectively scored readiness provides no reproducible predictive improvement over case-size, duration, and evidence-completeness baselines, or repeatedly assigns high readiness to unresolvable critical failures.

H2 — Handoff propagation. **2a: Evidence-bearing, structured handoffs reduce total human work across the connected process.** Production, verification, reconciliation, and repeat work all enter the endpoint; the predicted format contrast is $\beta_F^{(h)} < 0$ for total hours (Section 17). **2b: At equivalent downstream benefit and quality, agents produce and verify the structured handoff at lower resource cost than humans.** The agent contrast is $\beta_A^{(c)} + \beta_{AF}^{(c)} < 0$ for upstream production-and-verification cost. This is the specifically agentic increment, separate from the value of structure.

Failure criteria: 2a fails without lower total hours; 2b fails without lower upstream cost and equivalent downstream benefit and quality. Both comparisons retain all assigned cases, including rejected artifacts and work transferred upstream.

H3 — Labor concentration in the human residual. As agents absorb the standard path, residual cases retain a larger fraction of baseline labor than their fraction of case volume. A small human queue can therefore contain a large share of the work. The prediction concerns baseline effort measured before deployment, not a difficulty score defined after observing the routing decision.

Failure criterion: the residual does not become enriched in baseline effort under the tested standard-case rollout, or this pattern disappears across held-out case classes. Random-audit and mandatory-signature queues provide comparison groups.

H4 — Residual workload prediction. Exception-adjusted forecasts predict human-capacity change better than the uniform-effort model $q + (1 - q)m$. The comparator gives every exception the baseline average effort and sets review to fraction m of that average. The component model additionally represents baseline selection, post-deployment exception effort, rework, and recurring human upkeep. Beating a comparator that omits hours actually worked would prove little, so a second comparator receives the same rework and upkeep hours and the directly estimated conditional means h_E and h_R . With coherent inputs it is algebraically identical to the component model; the separation of κ from η earns its keep only when the routing rule, the case mix, or the agent configuration changes, because ϕ can then be re-estimated on historical work while η and μ are carried over. H4 therefore tests a transferability hypothesis: that effort ratios carry over to new conditions better than conditional hours do. It is plausible where assistance scales with the baseline difficulty of a case, and doubtful where it helps some categories of exception much more than others, since a change of mix then changes the average η without any change in the agent.

Failure criterion: using prospectively estimated inputs, the component model does not improve held-out total-hours forecasts over the uniform comparator and a stratified case-mix baseline, does not beat the matched comparator after a change of routing, mix, or configuration, or exceeds a prespecified error tolerance.

H5 — Delegation before authorization. At consequential gates, delegation of analytical production advances ahead of delegation of final authorization. The enterprise changes who prepares and executes a decision before changing who is empowered to approve it and answer for its consequences. H5 describes the *order* of the transition; Law 2 states its endpoint.

Failure criterion: analysis and authorization transfer at the same rate, or authorization transfers earlier, in settings where both can vary. A universal human-signature mandate explains a fixed boundary but cannot by itself validate the prediction. The lag is an institutional outcome to measure, not a claim about a uniquely human cognitive capability (Novelli et al., 2024).

H6 — Residual exhaustion. The set of indispensable human interventions shrinks toward empty: categories of intervention close faster than new ones appear, including those created by supervising the agents. H1–H5 can all hold while a small team remains indispensable. Three propositions of increasing strength must not be confused: one human intervention can be automated; the remaining interventions become automatable one after another; and there is a date after which no necessary human intervention remains. The law needs the third. H6 is the hypothesis that separates complete substitution from durable complementarity. Each remaining intervention is recorded by category and by origin: present at baseline, displaced from another role or organization, or created by the automation itself.

Failure criterion: over three successive annual evaluations, the number of indispensable intervention categories, or the hours they absorb, stops falling, or newly created categories offset those that close. Such a result rejects this trajectory hypothesis; like the plateau of Section 17, it is

warning evidence for the endpoint, not a finite-date refutation of complete substitution. It can coexist with the separate 80% FTE reduction.

9 Three worked routes and their substitutable transformations

Each generic route is illustrated by one case from the practitioner framework. The initial situations and the recorded opinions come from that framework; the route orders are the illustrative configuration of Figure 1 (Section 2.1), and the proposed machine operations are the implementation examined by the thesis. The cases are not reports of tested autonomous deployments.

9.1 W1, Buy: a new platform, from supplier request to the general gate

The framework’s SaaS CRM serves 2,000 sales users and handles confidential customer data. It plans SSO with optional MFA, lacks contractual SIEM log export, and has untested restoration with no defined recovery-time objective. Its Security opinion is red, with log export to be negotiated before signature and MFA and recovery conditions before go-live (Canale, 2026, p. 103). The same dossier gives work to every gate: Procurement and Legal carry the negotiation of the log-export clause, Compliance qualifies the customer data, IT checks the catalog fit and the run model, Finance validates the cost, and the general gate arbitrates. The Security opinion is one transformation among them.

Table 9: W1, Buy, in the illustrative order. The proposed machine operations preserve the CRM case’s initial red opinion; they do not assert a later approval.

Gate	Input	Output and automation target
Procurement	Need, budget, requested controls, supplier proposals and evidence.	Sourcing questions, offer comparison, evidence requests, due-diligence result.
Legal	Draft contract, service-level and data annexes.	Redlined clauses: log export, audit right, liability, reversibility.
Compliance	Data categories, purposes, hosting locations.	Processing assessment, transfer and retention conditions.
Security	Supplier security package, flows, identity model, classification.	Evidence-backed opinion; contractual changes and preconditions, including the MFA and recovery conditions.
IT	Catalog, run and support model.	Catalog fit, run and support model, operating constraints.
General	Opinions, reservations, business case, budget, owners.	Go, go with reservations, rework, or no-go, with owners and dates.

In this configuration the enterprise consults Security before the run model is fixed; automation must respect the order the enterprise has chosen and must not send a technically integrated platform to Security after the fact. The reusable object is the evidence record of Section 4.2, here a contract with exact clause locators; Legal, Compliance, and Security reuse it instead of re-extracting the same PDF. An automated red disposition is a *successful* execution of this example when the reference requirements still fail. Automatically producing a green status would not be substitution; it would be degradation of the governance function. The supplier belongs to another organization: buyer-side automation cannot hide the supplier’s evidence-production labor, and the complete claim for this route includes that interaction.

9.2 W2, Integrate: an information system inherited from an acquisition

The acquired entity has 400 people, its own on-premises Active Directory, cloud mail, and a provider-hosted ERP. A payroll requirement motivates network interconnection within thirty

days, despite missing account inventories, no endpoint detection and response (EDR), local logs, and an undocumented past incident (Canale, 2026, p. 104). IT first frames integration, isolation, or retirement; Architecture and Security then assess that scope, Legal and Procurement recover the inherited ERP obligations, Compliance qualifies the inherited data, the general gate’s support staff track the dated obligations, and Finance prices the integration options. The corporate acquisition decision itself is outside this route.

Table 10: W2, Integrate, in the illustrative order.

Gate	Evidence received	Work, output, and automation target
IT	Payroll need; inherited AD, mail, ERP; available inventories and provider documents.	Frame what will be integrated, isolated, or retired; plan the run takeover; transmit scope, dependencies, and explicit evidence gaps.
Architecture	IT scope, inherited flows and interfaces, network constraints.	Target integration architecture; design of the restricted connection; conditions.
Security architecture	Scope; identity, exposure, incident, EDR, and logging evidence.	Assess integration risk; carry the framework’s conditional (yellow) opinion into a risk card, a restricted connection scope, and dated control requirements.
Legal	Hosted ERP contract, inherited service obligations.	Obligations register; novation or exit conditions.
Compliance General	Inherited personal data, applicable rules. Gate opinions, residual risks, integration owner, cost and schedule.	Conditions on data access and retention. Record the decision and its authorized scope, accepted residual risk, conditions, and follow-up; preserve the integration director’s risk ownership.

The framework’s path overview prohibits interconnection before the acquisition audit. Its worked case records a bounded exception: a filtered partner-zone connection during the audit without joining directories, EDR before any connection, the account inventory within thirty days, SIEM log integration within sixty days, and a risk card signed by the integration director; directories are joined only after the audit (Canale, 2026, pp. 18, 104). These are different prerequisites and dated obligations, not a blanket approval.

An agent can reconcile inventory exports, link accounts to owners, locate provider reports, and maintain the reservations register that later gates consume instead of reconstructing the audit. The model of Section 10 makes the remaining work explicit: investigating the undocumented incident concentrates baseline effort in exceptions (κ , hence ϕ); pursuing unavailable provider reports determines how much effort persists after assistance (η); recurring condition checks consume standard-path review (μ); and maintaining integration playbooks consumes fixed upkeep (B_A). A missing report remains missing: an agent must obtain evidence, run an accepted alternative investigation, or issue a justified unresolved or refusal disposition. A complete-looking record does not complete an audit. Low repetition, inherited state, and supplier dependence make Integrate the most demanding test of the law; shared provision addresses scale, not those information deficits.

9.3 W3, Build: a new internal project

The third case places the framework’s internal IT-support agent on the Build route (Canale, 2026, pp. 19, 105). The proposed service uses Copilot Studio and Microsoft Foundry (formerly Azure AI Foundry; Microsoft, 2026), reads tickets and SharePoint through its integrations, and accesses Microsoft Graph. It uses the developer’s identity and service principal, can reset passwords without approval, lacks prompt and action logs, and retrieves across the whole SharePoint index.

Tickets contain personal data. The framework rejects this design with a red opinion and proposes a read-only knowledge-base pilot without Graph as an alternative. Around that opinion, IT checks the catalog and the run model, architects frame the target design, readiness experts review tests and runbooks, the general gate’s support staff and Finance assemble schedule and budget, and sponsors commit at the general gate.

Table 11: W3, Build, in the illustrative order. Rows after the Security opinion describe required reviews of a revised design, not decisions already obtained.

Gate	Evidence received	Work, output, and automation target
IT	Support need; proposed service and platforms, costs, operating constraints.	Catalog conformity, run and support model, outstanding questions.
Architecture	Design, integrations with tickets, SharePoint, and Graph, data flows.	Target architecture; decisions on interfaces and retrieval scope.
Security architecture	Permissions, identities, retrieval scope, logging evidence; ticket data.	Issue the red opinion. Specify a dedicated identity, controlled actions, permission-filtered retrieval, and logs; route redesign or the read-only pilot back for review.
Tech readiness	Revised design after re-review; pilot and test results; runbooks.	Check conditions against actual access and action tests; produce readiness evidence and unresolved items.
General	Gate opinions, residual risks with business owners, budget, schedule, mandate.	Record the decision for the defined scope and its conditions, and the sponsors’ commitment. Generating the dossier does not supply that authorization.

Automation can assemble the dossier, compare the declared read-only scope with actual permissions, and link each condition to a test and a responsible owner. Difficult identity defects contribute to κ ; assistance with their resolution determines η ; checking corrected conditions contributes to μ ; repeated failed reviews contribute to rework; maintaining permission tests contributes to B_A . Corrected evidence and authorized decisions, not generated documentation, clear the red opinion.

The agent being evaluated and the system executing the DGF are separate principals, with separate identities and authorities. A project cannot approve itself by generating its own evidence. A fully automated arrangement preserves separation of duties through independent identities, evidence sources, execution restrictions, and authorization policies; it need not identify separation of duties with two human bodies. The sponsors’ commitment at the general gate is deliberately not removed from the thesis: it becomes the exercise of a standing mandate (Section 7.3). Reusable patterns and testable environments make Build the most plausible first route to close.

10 The instrument: a complete labor account and the workload frontier

10.1 Count the complete human workload

Compare a human operating regime H with an agent-assisted regime A . For one work package, let λ be visits per period and $h_0 > 0$ mean baseline case-handling hours. In the agent regime, q is the fraction requiring substantive human exception handling; h_E is the complete human time on an exception; h_R is mean human review on a non-exception case; and h_W is additional rework per visit. Fixed, case-independent human upkeep is B_H or B_A .

$$\bar{h}_A = qh_E + (1 - q)h_R + h_W, \quad (4)$$

$$L_H = \lambda_H h_0 + B_H, \quad L_A = \lambda_A \bar{h}_A + B_A, \quad (5)$$

$$N_s = L_s / K_s, \quad s \in \{H, A\}. \quad (6)$$

N_s is workload-equivalent FTE at K_s useful hours per period. Exception effort includes any preliminary human review on those same cases. h_R includes the expected cost of sampled audits and mandatory approvals on the ordinary path. h_W excludes work already attributed to these terms or to repeat visits in λ . B_A includes policy maintenance, agent evaluation, training, and incident review within the comparison boundary. Shared teams are allocated once. The account is invariant to the way an enterprise partitions its gates: the same activities, hours, and costs give the same total whether they are called one gate or three, so splitting or merging gates cannot create or hide labor.

This boundary follows the business process rather than a single department's dashboard. A system that reduces one team's reading time but transfers more remediation to project engineers has not necessarily reduced enterprise work. Likewise, a human waiting exclusively on an agent consumes capacity; inference that runs without human attention does not.

10.2 Separate selection from post-deployment effort

A high exception cost has two different explanations: the routed cases were expensive already, or deployment made them more expensive. Let h_E^H and h_S^H be their conditional baseline means under a frozen case partition. Define

$$\kappa = \frac{h_E^H}{h_0}, \quad \sigma = \frac{h_S^H}{h_0}, \quad \eta = \frac{h_E}{h_E^H}, \quad k = \kappa\eta. \quad (7)$$

Here κ measures baseline selection; η measures post-deployment effort inflation or assistance for that group. For $0 < q < 1$ and positive h_E^H ,

$$q\kappa + (1 - q)\sigma = 1, \quad \sigma = \frac{1 - q\kappa}{1 - q} \geq 0. \quad (8)$$

Consequently $q\kappa \leq 1$. This constraint applies to κ , not to the post-deployment multiplier k . The same baseline exception group can take longer after deployment, so qk may exceed one. A fixed routing rule can be applied in shadow mode to historical cases to estimate this partition; changing the rule requires a new partition or explicit reweighting.

Proposition 1 (Residual selection and retained baseline labor). *Let $T \geq 0$ denote integrable baseline handling time with $\mathbb{E}[T] = h_0 > 0$. Let $J \in \{0, 1\}$ indicate assignment to the human residual and $e(T) = \Pr(J = 1 \mid T)$, with $q = \mathbb{E}[e(T)] > 0$. Then*

$$\mathbb{E}[T \mid J = 1] - h_0 = \frac{\text{Cov}(T, e(T))}{q}, \quad (9)$$

$$\phi := \frac{\mathbb{E}[TJ]}{h_0} = q\kappa = q + \frac{\text{Cov}(T, e(T))}{h_0}. \quad (10)$$

Thus the residual's baseline labor share exceeds its case share exactly when routing is positively associated with baseline effort. If e is nondecreasing, this covariance is nonnegative. No ordering of conditional variances follows.

Proof. The law of iterated expectations gives $\mathbb{E}[TJ] = \mathbb{E}[Te(T)]$. Subtracting h_0q and dividing by q gives Equation (9); dividing by h_0 instead gives Equation (10). For independent copies T, T' , the covariance has the representation

$$2 \text{Cov}(T, e(T)) = \mathbb{E}[(T - T')(e(T) - e(T'))].$$

A nondecreasing e makes the integrand nonnegative. The identities involve means and a cross-moment, not a constraint ordering the variances of T and $T \mid J = 1$. $\square$

The economically important quantity is ϕ , not q alone. A residual containing 20% of cases can retain 40% of baseline labor when $\kappa = 2$. With post-deployment inflation $\eta = 1.5$, that group's effort becomes 60% of the entire baseline case workload before review, rework, and upkeep are counted. The selection identity supplies the bridge between exception routing and the capacity calculation. The empirical question in Hypothesis H3 is whether actual routing produces the positive covariance.

10.3 Compression versus expansion

For positive baseline means in both groups, define $\mu = h_R/h_S^H$: the fraction of the standard group's baseline effort required for review. Like η for exceptions, μ measures post-deployment effort against that group's own baseline. Retain $m = h_R/h_0 = \sigma\mu$ for comparisons in overall-mean units, and set $r = h_W/h_0$, $b_s = B_s/(\lambda h_0)$. At common visits and useful hours,

$$\rho := \frac{N_A}{N_H} = \frac{q\kappa\eta + (1-q)\sigma\mu + r + b_A}{1 + b_H} = \frac{\phi\eta + (1-\phi)\mu + r + b_A}{1 + b_H}. \quad (11)$$

The second form reads directly. The share of baseline labor held by the exception group is multiplied by the change in its effort; the share held by the standard group is multiplied by the fraction of that effort still spent on review; rework and upkeep are added. ϕ can be measured on historical work before any deployment, whereas η and μ are what the deployment determines.

If $h_S^H = 0$, μ is undefined and the dimensional form (4) applies. Effort ratios can exceed one; $q\kappa \leq 1$ remains the baseline partition constraint. For fixed $k = \kappa\eta > m$ and fixed assurance ratios, the equivalent exception-share threshold is

$$q < q^* = \frac{1 + b_H - m - r - b_A}{k - m}. \quad (12)$$

A negative q^* allows no saving at nonnegative q ; a value above one places the boundary beyond the feasible case-share range. If $k = m$, the case share has no direct effect. If $k < m$, the inequality reverses. These are algebraic boundaries, not fitted empirical thresholds.

The same boundary can be read as a margin on exception effort. For $\phi > 0$, workload stays below the baseline only while

$$\eta < \eta^* = \frac{1 + b_H - (1 - \phi)\mu - r - b_A}{\phi} = \mu + \frac{1 + b_H - \mu - r - b_A}{\phi}. \quad (13)$$

Whenever the standard path alone would save labor ($\mu + r + b_A < 1 + b_H$), the larger the residual's share of baseline labor, the smaller this margin.

At $q > 0$, the boundary is $k^* = m + (1 + b_H - m - r - b_A)/q$. Figure 3 draws it separately for each route: fixed upkeep per visit differs with volume, so a single curve cannot classify both. Workload increases above the applicable curve. A lower escalation rate can thus be outweighed by costly exception resolution or assurance.

The same equation states what the law of gate substitution requires. Complete substitution drives each weighted contribution in the numerator of Equation (11) to zero: exception labor, standard-path labor, rework, and support. The law predicts that endpoint; the equation measures the distance to it, term by term, and names the term that blocks each route. Tracking the origin of each residual term over time, which interventions disappeared, which moved, and which are new, is what Hypothesis H6 tests.

10.4 Workload, staffing, and cash are distinct outcomes

A workload-equivalent FTE is an effort unit. Staff needed to meet deadlines also depend on arrival variability, specialization, minimum coverage, and availability of approvers. A utilization constraint $L \leq unK$, with target $u < 1$ and staff n , is only a starting point; a delay target needs a queueing or scheduling model. The present calculations hold useful-hour conversion constant, not a measured waiting-time distribution.

Similarly, saved hours are not automatically avoidable salaries: the component model identifies a resource change, and a financial counterfactual determines which part becomes cash or additional output. Section 12 derives the staffing consequence.

11 A 140-FTE enterprise: identical coverage, opposite outcomes

11.1 A coherent baseline and four operating regimes

Take the nine populations in Table 3, full useful-time allocation to the modeled activities, 100 aggregated cases per population per month, and $K = 120$ useful hours per FTE. Their counts then equal baseline workload-equivalent FTE. Each population’s mean handling time is $h_0 = 120N_0/100$. Ordinary baseline support work is inside these aggregate case times; the example has no separately recorded baseline upkeep B_H . Approval work is included among the relevant review activities.

Each resource envelope aggregates a population’s contributions across the three routes. The 100-case envelope is a pooled accounting volume, not 100 programs per month or 140 FTE assigned to each route. Route-specific case counts, effort, and routing parameters require separate measurement; shared upkeep is allocated once. No split between routes is assumed in these sensitivity totals; Section 11.3 isolates two routes for one specialist team, and Appendix D allocates the pool across all three.

Exception fractions q range from 0.16 to 0.33. Set $\kappa = 2$ for every population. Equation (8) then determines each standard-group baseline mean. For the 33% groups, $\sigma = 0.5075$; for the 16% group, $\sigma = 0.8095$. These positive conditional means define the same baseline across all four regimes.

Table 12 changes η , review, rework, and upkeep while keeping q and κ fixed. S0 halves human effort on the baseline exception group, giving $k = 1$; S3 raises it by 50%, giving $k = 3$. Table 12 holds m common across populations, so the group-relative review ratio $\mu = m/\sigma$ varies. Table 30 reports both normalizations without changing the scenario inputs.

Table 12: Synthetic operating assumptions. $\kappa = 2$ and the case partition are common. h_W is additional hours per case; B_A is hours per population per month.

Regime	η	$k = \kappa\eta$	m	h_W	B_A	Interpretation
S0	0.50	1	0.10	0	0	Assisted exceptions; low recurring overhead.
S1	1.00	2	0.10	0	120	Baseline exception effort retained; human upkeep added.
S2	1.00	2	0.10	0.50	120	S1 with additional case rework.
S3	1.50	3	0.35	2.00	240	Inflated exceptions and heavier assurance.

11.2 Where the work goes

The largest population illustrates the mechanism; every row of Table 13 reads the same way. For Architecture, $h_0 = 36$ hours, $q = 0.18$, and $\kappa = 2$ imply baseline exception handling of 72 hours and baseline standard-case handling of 28.10 hours. Under S0, exceptions require 36 human-hours

Table 13: Calculated workload-equivalent FTE at fixed synthetic volumes and coverage. Totals use unrounded values; population identifiers match Table 3.

Pool	Baseline	h_0 (h)	q	S0	S1	S2	S3
R1	10	12.00	0.22	2.98	6.18	6.60	13.00
R2	10	12.00	0.33	3.97	8.27	8.69	15.91
R3	30	36.00	0.18	7.86	14.26	14.68	28.48
R4	20	24.00	0.16	4.88	9.08	9.50	19.15
R5	10	12.00	0.22	2.98	6.18	6.60	13.00
R6	25	30.00	0.24	7.90	14.90	15.32	28.32
R7	15	18.00	0.26	5.01	9.91	10.33	19.25
R8	10	12.00	0.33	3.97	8.27	8.69	15.91
R9	10	12.00	0.33	3.97	8.27	8.69	15.91
Total	140	–	–	43.52	85.32	89.07	168.92

after assistance and standard-path review 3.6 hours. This yields 7.86 workload-equivalent FTE out of 30. Under S1, exception effort remains 72 hours: direct case labor is 13.26 FTE, and 120 monthly upkeep hours bring the total to 14.26. S2 adds 0.5 hours per case and yields 14.68 FTE.

For all nine populations, totals are 43.52, 85.32, 89.07, and 168.92 FTE. Figure 2 shows the components. Weighting populations by baseline workload, the exception fraction averages 23.43%; the frozen exception groups retain 46.86% of total baseline case labor. Multiplying that retained labor by different η values is a major source of the difference across operating regimes. Review and fixed upkeep complete the comparison.

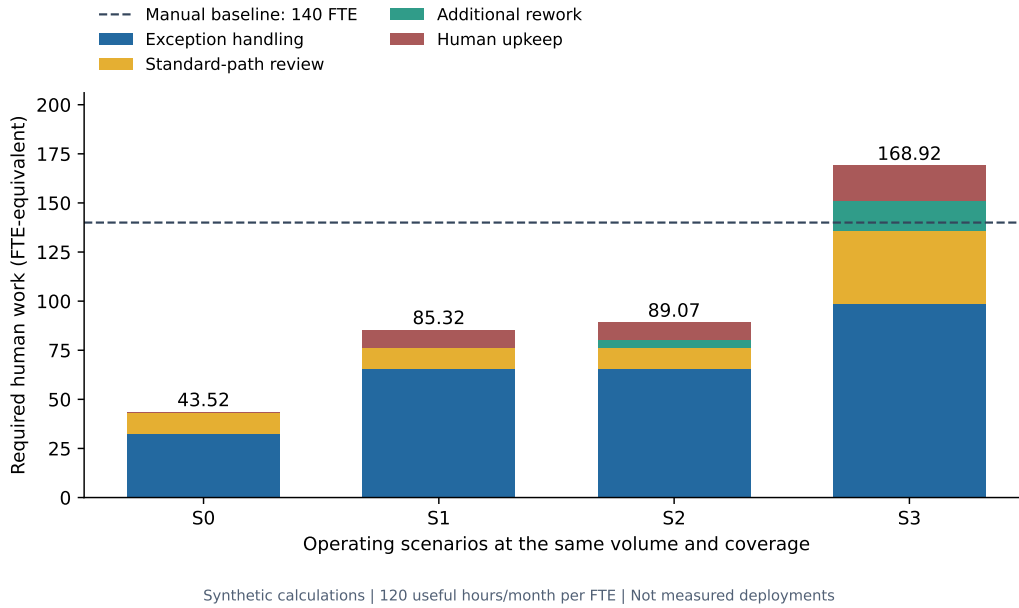


Figure 2: The same coverage can release or consume human capacity. Synthetic totals decompose into exception handling, ordinary-path review, rework, and fixed upkeep. S0–S3 use the same case volumes, exception fractions, and baseline selection; their post-deployment burdens differ.

Equation (13) turns the same inputs into a fragility margin, reported in the last column of Table 30. Holding the S2 review, rework, and upkeep fixed, the exception group could take 2.64 times its baseline effort in Security Architecture, 2.42 in Architecture, 1.81 in IT Production, 1.77 in Procurement and in Compliance, and 1.60 in Tech Readiness before the saving disappears. For Legal, General Gate Support, and Finance, whose residual holds two-thirds of baseline

labor, the margin is 1.20: a rise of about 20% in exception effort erases the entire gain. The populations with the largest exception share are the most exposed to post-deployment inflation.

These scenarios answer a conditional question: how much human work would the process require at the specified operating burdens? They are not a probability distribution over future outcomes. The notable result is not the most optimistic number but the failure of coverage to identify which side of the frontier the process occupies. In an enterprise evaluation, the component quantities are estimated prospectively and the forecast is tested on subsequent total hours.

11.3 Same platform, different route economics

The DGF routes differ in volume as much as in content. Integrations of inherited systems are rare and heterogeneous; internal builds are frequent and more alike. Table 14 takes one specialist team of 20 FTE with 100 monthly review visits and splits them into five Integrate visits and 95 Build visits, each with $h_0 = 24$ hours. Both routes use the same platform, 2.4 review hours and 0.5 rework hours per visit, and each carries 60 shareable plus 6 local upkeep hours per month: the playbooks, routing rules, and agent evaluations of Section 9.2. The Integrate route escalates 60% of its visits, with $\kappa = 1.4$ and unchanged exception effort; the Build route escalates 14%, with $\kappa = 2$, and assistance lowers exception effort by a fifth. A visit is a review work package, not a whole integration program. Both partitions are coherent: $q\kappa = 0.84$ for Integrate and 0.28 for Build, and Integrate’s baseline standard cases still cost 9.6 hours. The assumptions are synthetic, not measurements of the framework’s cases.

Table 14: Route-specific illustration for one specialist team. Fixed upkeep is the same number of hours on both routes, but its burden per visit differs.

Route	Visits	q	κ	ϕ	η	Upkeep h/visit	b_A	ρ
Integrate (W2)	5	0.60	1.4	0.84	1.0	13.20	0.550	1.451
Build (W3)	95	0.14	2.0	0.28	0.8	0.69	0.029	0.360

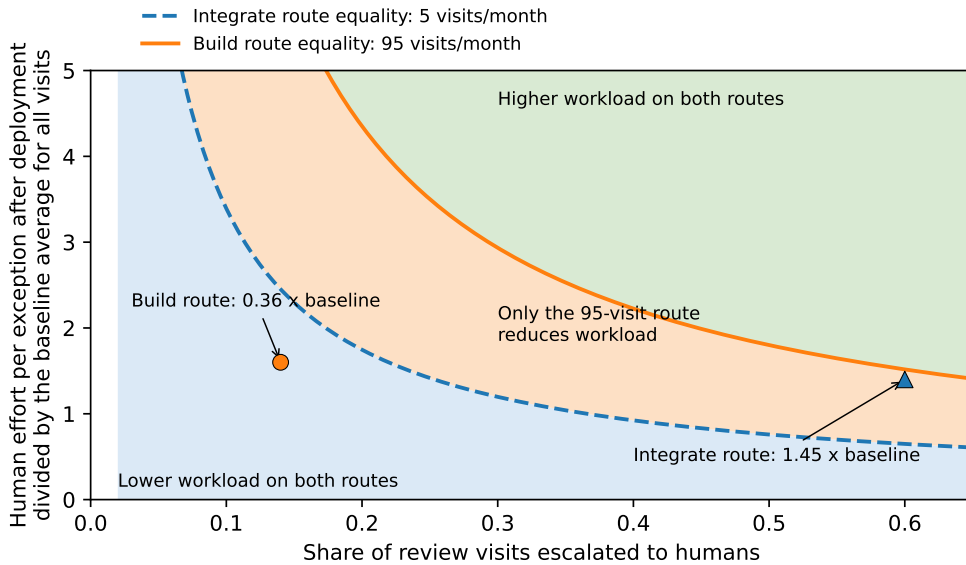


Figure 3: One platform, two workload frontiers. Each route has 24 baseline hours per visit, review at 10% of that mean, 30 minutes of rework per visit, 60 shareable plus 6 local monthly upkeep hours, and no separate baseline upkeep. The Integrate route has five visits per month; the Build route has 95. Above a route’s curve, its human workload exceeds its baseline. Markers show the two synthetic routes of Table 14; volumes count review visits, not programs.

The ratios are 1.451 and 0.360: opposite workload outcomes on the same platform (Figure 3). The Integrate route combines concentrated residual labor with 13.2 upkeep hours per visit, versus 0.69 on the Build route. Holding Integrate’s other settings fixed but increasing its volume to 95 visits lowers its ratio to 0.930: scale alone crosses its frontier. Sharing the sixty common hours across 1,000 customers lowers it to 0.951 (Table 8). A deployment that would be a saving at scale is a loss at five visits a month.

The pooled ratio of 0.41 hides this: 8.29 residual FTE against 20. An enterprise reading only the aggregate would miss that one route consumes more specialist time than before. These route curves and points replace any presumption that one platform has one economic result.

12 Workforce consequences: fewer FTE for the same governed output

The purpose of automating a gate is not a faster document. It is that the same governed output requires fewer people. This section states that consequence without euphemism and marks exactly what it depends on.

12.1 From vanishing human effort to vanishing posts

Composition. If each component satisfies its contract on every valid history it can receive, outputs satisfy successor input requirements, and the controller and recovery are also automated, the route executes without a human step. This is ordinary composition of compatible procedures. With at most M visits and a valid-prefix conditional failure bound ϵ , summing first-failure events gives a route bound $M\epsilon$, without an independence assumption. An average benchmark result supplies neither that conditional bound nor a guarantee on arbitrary new inputs.

Labor. For a finite registry of packages, let $\nu_i(t)$ be visits, $\ell_i(t)$ mean human hours per visit, and $B(t)$ nonduplicated recurring support. Then

$$L(t) = \sum_i \nu_i(t)\ell_i(t) + B(t), \quad N(t) = L(t)/K. \quad (14)$$

At fixed output demand with bounded visits, $\ell_i(t) \rightarrow 0$ and $B(t) \rightarrow B_\infty$ imply $N(t) \rightarrow B_\infty/K$. If support is also automated, $B_\infty = 0$ and the requirement tends to zero. The arithmetic consequence is direct: the same completed work requires fewer workers’ hours, and complete closure removes the execution labor requirement. A declining share of escalated cases is not enough: $q(t) = 1/t$ with exception hours $h_E(t) = t$ leaves expected exception labor at one hour per case. Exact zero staffed execution requires removal of every necessary human task, not rounding a small FTE to zero; a positive ten-minute obligation can still require an available person.

Posts. With interchangeable workers, useful capacity uK per person at target utilization $0 < u < 1$, positive wage cost, and fixed nonlabor costs, cost-minimizing headcount is

$$n^*(t) = \left\lceil \frac{L(t)}{uK} \right\rceil. \quad (15)$$

For $n_0^* = \lceil L_0/(uK) \rceil$, a reduction of at least one post occurs exactly when $L(t) \leq (n_0^* - 1)uK$. At complete closure $L = 0$ and $n^* = 0$. Reduced staffing is thus a direct implication once productive necessity and the staffing objective are specified. Service levels and heterogeneous skills add the requirements of Section 10.4.

An employer can retain people, shorten working time, or assign new work. Those choices do not restore demand for the displaced gate execution. If the organization no longer needs ten people

to operate the same Procurement package, employing those ten elsewhere is a new allocation of labor, not evidence that the old gate still needs them. The direct conclusion is a reduction of the workers required by the replaced workflows; actual workforce reductions follow when staffing adjusts instead of the released capacity being absorbed by new demand. In the author's experience, that adjustment is a stated objective of some governance automation programs.

12.2 The dated prediction: 80% fewer required FTE by 2033

The author's dated hypothesis is that, by 20 September 2033, the DGF ecosystem will require at most 20% of its 2026 workload-equivalent FTE for comparable governed output, quality, and service. FTE means full-time equivalent: a unit of human work, not a count of individual people. The model already defines $N(t) = L(t)/K$ in Equation (14); its examples use $K = 120$ useful hours per month. Part-time contributions are added as fractions, and supervision, exceptions, maintenance, and supplier support remain in the human-work account.

Applied to the illustrative 140-FTE baseline, the threshold is $140 \times 0.20 = 28$ required FTE. Figure 8 places this 2033 hypothesis beside the existing configurations: 140, 60.56, 16.35, 5, and 0 FTE. These are conditional configurations, not successive dated observations. The 60.56-FTE configuration exceeds the threshold, whereas the 16.35-FTE configuration falls below it. Five support FTE count inside the 28-FTE ceiling, leaving at most 23 FTE for other work. The illustrative pool is not a census of the whole ecosystem; empirical supplier work must be added consistently to both the baseline and follow-up boundary.

This is a proposed FTE hypothesis, not a benchmark result or a prediction of 80% fewer individual employees. Actual staffing also depends on utilization, skills, service constraints, and redeployment. An 80% reduction in required FTE allows continuing human execution and support; complete automation remains a stronger conjecture without a fixed date.

12.3 The trajectory and its support floor

Appendix D allocates the 140 FTE of Table 3 across the three routes and computes illustrative substitution configurations. With 30%, 75%, and 25% of baseline route labor remaining on W1, W2, and W3 and five FTE of shared support, the requirement is 60.56 FTE; with 5%, 20%, and 3%, it is 16.35; with direct gate execution closed, the five support FTE remain, a reduction of 96.4%; with support closed, it is zero. These are chosen fractions, not an estimated trajectory: they show what the account returns, not when.

A fixed support requirement prevents a zero-worker endpoint even after every direct case activity is automated. Law 5 predicts substitution of this remaining work as well; the model does not make the floor vanish by renaming its employees supervisors. Conversely, a small supervisor team is already a large structural reduction from a large case-processing population. Section 11 shows the other side of the same account: before the endpoint, badly governed automation can *raise* the requirement above 140.

12.4 Employment beyond the DGF

Acemoglu and Restrepo (2019) distinguish displacement from new-task reinstatement and productivity effects. Those mechanisms can offset losses in particular activities. If output demand scales by $g(t)$ while labor per output is $h(t)$, total variable labor follows $g(t)h(t)$: halving labor intensity while doubling demand leaves it unchanged. New work on agent governance can add to $B(t)$ or create a genuinely new output class. Including it is not softening displacement; it is measuring whether the displaced labor is actually replaced by another demand.

A permanently positive support floor defeats the stronger zero-labor closure even if the original

review workforce is much smaller. The paper does not equate either outcome with an automatic economy-wide reduction in employment; it does claim that the governance specialists of the DGF are among the first white-collar populations for whom the question stops being hypothetical.

13 How handoff effects travel through the enterprise

For a dependency $i \rightarrow j$, let z measure a prespecified feature of the artifact supplied upstream: for example, the share of required claims with independent evidence, a usable source locator, and a version reference. Formatting and semantic adequacy are measured separately. Case mix x , agent configuration θ , and policies Π also matter. The response functions of interest are

$$(q_j, h_{Ej}, h_{Rj}, h_{Wj}, B_j) = f_j(z, x, \theta, \Pi). \quad (16)$$

At fixed visits, the total labor per visit including allocated upkeep is

$$g_j(z) = q_j h_{Ej} + (1 - q_j) h_{Rj} + h_{Wj} + B_j / \lambda_j.$$

Its derivative decomposes the channels:

$$\frac{dg_j}{dz} = q'_j(h_{Ej} - h_{Rj}) + q_j h'_{Ej} + (1 - q_j) h'_{Rj} + h'_{Wj} + B'_j / \lambda_j. \quad (17)$$

In the Integrate example, a verified inherited-system inventory can resolve missing facts before Architecture and Security repeat the search. In the Build example, a permission-and-test record can carry a security condition into readiness checks. Such handoffs can lower q_j , h_R , or h_W through fewer unresolved claims, faster verification, or fewer inconsistent versions. Conversely, a new signature-verification requirement can raise upkeep; an apparently complete but false record can cause downstream repair. The empirical effect is the sum of these channels. If visit counts change, total labor additionally changes through the frequency of passage and return.

The positive cascade hypothesis concerns these response functions, not a scalar bonus attached to an arbitrary index. A gate with no authorized access to a required system does not acquire it because its input is well formatted. Appendix B provides a bounded representation that preserves such blockers.

Hypothesis H2 separates the two sources of gain (Section 17). If a typed human record and a typed agent record generate the same downstream benefit but the latter is cheaper to produce and verify, there is an incremental agentic gain through upstream production. If both cost the same, there is a valuable interface redesign but no demonstrated agentic increment. If the agent's record requires more verification, the human-structured workflow may be economically superior. This is a comparison among implementations of the same coordination function.

14 From workload to enterprise economics

14.1 Comparable resource costs

Let w_s be a human-hour valuation, a_s variable nonlabor cost per visit, F_s fixed nonlabor cost, and ℓ_s expected attributed error loss per visit under regime s . A common-boundary average cost is

$$\bar{c}_s = w_s \bar{h}_s + a_s + \frac{w_s B_s + F_s}{\lambda_s} + \ell_s. \quad (18)$$

Human upkeep is included through B_s , not charged again in F_s . Both regimes bear error losses. Different w_s values allow a more highly paid specialist residual. For mixed roles, the labor valuation becomes a sum of hours times role-specific costs.

For the IT Production and Standards population under S2 ($h_0 = 30$ hours, $q = 0.24$), $\bar{h}_A = 17.18$ and $B_A/\lambda = 1.2$ hours. Take artificial monetary units with $w_H = 80$, $w_A = 100$, $a_H = 20$, $a_A = 50$, $F_H = 1000$, $F_A = 4000$, and $\ell_H = 30$, $\ell_A = 50$ at 100 monthly visits. Then

$$\bar{c}_H = 2460.00, \quad \bar{c}_A = 1978.00. \quad (19)$$

Holding the other terms fixed, $\ell_A = 532.00$ is the error-loss value at which the costs equalize. A separate security or safety constraint can still prohibit deployment below this financial threshold. The numbers illustrate opportunity costs, not market salaries, purchase prices, or payroll reductions.

14.2 Average cost, scale, and system return

When variable cost $v_s = w_s \bar{h}_s + a_s + \ell_s$ is constant and fixed cost is $J_s = w_s B_s + F_s$, total period cost is $T_s(\lambda) = J_s + \lambda v_s$. Average cost is $J_s/\lambda + v_s$ and marginal cost is v_s . Fixed assurance becomes cheaper per case as volume grows, but capacity thresholds or congestion can change this relationship. A high-volume standardized route may justify an investment that is uneconomic for a rare project type; Section 11.3 gives the workload counterpart.

At pipeline level, a shared incident must be counted once. An aggregate comparison is

$$C_s = \sum_i [w_{si} L_{si} + \lambda_{si} a_{si} + F_{si}] + \mathbb{E}[\mathcal{L}_s], \quad (20)$$

where $\mathcal{L}_s$ is total process loss over the period. Shared labor and technology are allocated once; local losses are not added again to the aggregate loss. The comparison covers the same requested business outputs and includes any different numbers of visits required to produce them.

For an investment horizon $0, \dots, T$, let V_t be nonduplicated attributable monetary benefits, I_t incremental implementation and operating expenditure, and d the discount rate. One explicit ROI convention is

$$\text{ROI} = \frac{\sum_{t=0}^T (V_t - I_t)/(1+d)^t}{\sum_{t=0}^T I_t/(1+d)^t}, \quad \sum_{t=0}^T I_t/(1+d)^t > 0. \quad (21)$$

Benefits already net of a cost cannot subtract that same cost again. Resource savings and cash savings require distinct benefit series when salaries remain committed. Transition work, migration, training, and any redundancy costs belong to the investment comparison.

Upstream gains can reach several departments, but their returns are not automatically superadditive. If $B(S) = C(\emptyset) - C(S)$ is net saving for intervention set S , the interaction

$$B(\{u, v\}) - B(\{u\}) - B(\{v\}) \quad (22)$$

is positive only when the joint intervention saves more than the isolated gains combined. Shared information, integration costs, or overlapping work can make the interaction negative. This test is the economic basis for investigating upstream leverage, rather than assuming it from process order.

15 Dated milestones with their assumptions exposed

The evidence supports measuring progress, not extrapolating a 50% benchmark into a production approval date. We separate four clocks: length capability, reliability qualification, evaluation, and institutional change. Their parameters are explicit sensitivity inputs.

15.1 Length capability: what three doublings would mean

Use 8 May 2026 as a calendar anchor and choose $H_{80,0} = 3$ human-hours for a planning calculation, close to the 80% horizon reported for the best model at that date (METR, 2026a). Extending the historical seven-month doubling interval to the 80% trajectory follows the similar trend reported by Kwa et al. (2025); transferring it to gate work is a separate assumption. For target task length D ,

$$T_{\text{length}}(D) = d [\log_2(D/H_{80,0})]_+ \text{ months}, \quad d = 7. \quad (23)$$

Table 15: Length milestones from a chosen three-hour 80% anchor on 8 May 2026 and a seven-month doubling interval. These are scenario dates, not observed gate capabilities or qualified deployment dates.

Target hours	Doublings	Exact months	Calendar months	Scenario date
12	2.000	14.00	14	8 Jul 2027
24	3.000	21.00	21	8 Feb 2028
36	3.585	25.09	26	8 Jul 2028

For twenty-four hours, three doublings give twenty-one months. That is a length milestone at the selected 80% criterion, not completion of a gate with a 20% failure allowance. The formula can be recalculated with an independently baselined gate horizon; changing the anchor changes the date rather than the meaning of success. Figure 4 shows how slower assumptions move the milestone outward.

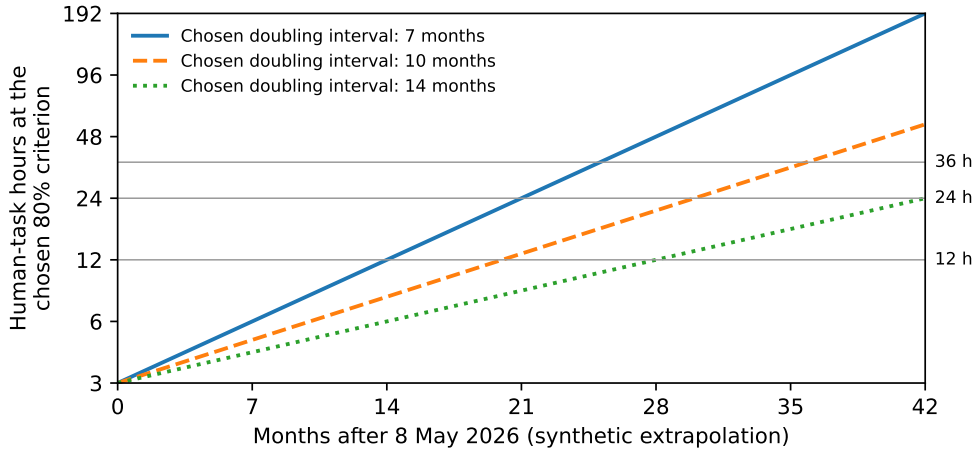


Figure 4: How the assumed pace changes the length milestone. All curves start from a chosen three-hour horizon at 80% success; none is a reconstructed METR series. A seven-month doubling reaches twenty-four hours in twenty-one months. Reliability qualification and institutional delegation are additional clocks.

15.2 The reliability tail has a separate clock and price

Let $e_0 = 0.20$ be the scenario's initial error probability and e^* a chosen qualification ceiling. The number of tenfold error reductions required is

$$z = \log_{10}(e_0/e^*), \quad T_{\text{tail}} = \nu z, \quad C_{\text{tail}}/C_0 = c^z. \quad (24)$$

Here ν is months per tenfold error reduction and c its multiplicative resource-cost factor. Neither is estimated from METR. Verification, retries, better evidence, or a better model could change the costs differently; a nonzero irreducible error floor makes the target unreachable.

Table 16: Reliability-tail sensitivity from initial error 0.20. Months per tenfold error reduction and cost factors are independent, unestimated assumptions. The cost multiplier is relative to C_0 , not a quoted price.

Error target	z	Tail months at $\nu =$			Cost multiplier at $c =$		
		6	12	24	2	4	10
10^{-3}	2.301	13.81	27.61	55.22	4.93	24.29	200.00
5×10^{-5}	3.602	21.61	43.22	86.45	12.14	147.45	4,000.00

The first row illustrates a 99.9% gate criterion. A route budget of 10^{-3} over twenty possible visits would instead require a conditional per-visit budget of 5×10^{-5} in the simple union-bound allocation; the second row shows how much more demanding that is. For a transparent serial planning scenario,

$$T_{\text{qualified}} = T_{\text{length}} + T_{\text{tail}} + T_{\text{evaluation}} + T_{\text{institution}}. \quad (25)$$

We use twelve months each for evaluation and institutional change only to illustrate their effect. Tasks can progress in parallel in reality, while legal or information constraints can delay them indefinitely.

Table 17: Illustrative qualification calendar for a 24-hour task and a 10^{-3} error ceiling. Each duration is rounded upward before calendar addition. Domain transfer, attainable error floors, and legal permission remain separate requirements.

ν	Length	Tail	Evaluation + institution	Total months	Scenario date
6	21	14	12 + 12	59	8 Apr 2031
12	21	28	12 + 12	73	8 Jun 2032
24	21	56	12 + 12	101	8 Oct 2034

These scenarios illustrate when individual work packages might qualify for automation. They do not derive the predicted 80% reduction in required FTE by 2033. That prediction also requires enough labor-intensive activities to change and new support not to offset the reduction. Work packages may remain human-operated while the FTE target is met.

For deployments evaluated throughout the final observation year of the proposed workforce study, 20 September 2032 is a useful planning cutoff, 76 whole calendar months and twelve days after the 8 May 2026 anchor. Conservatively using 76 months, qualification of this particular 24-hour work package requires $21 + \nu z + 24 \leq 76$, or $\nu \leq 31/z$: approximately 13.47 months per tenfold error reduction at a 10^{-3} ceiling, or 8.61 months at 5×10^{-5} . At the first ceiling, the April 2031 and June 2032 scenarios fit that window; October 2034 does not. These are work-package sensitivities, not necessary or sufficient conditions for an 80% FTE reduction. Complete execution and support closure would depend on the slowest contract and its shared dependencies; that stronger endpoint has no fixed date in this paper.

15.3 Route order is a hypothesis, not a relabelled trend

We propose testing $W3 \prec W1 \prec W2$, Build before Buy before Integrate, as an adoption-order hypothesis for comparable enterprises: builds can be designed to generate testable evidence; purchases add supplier negotiation and evidence dependencies; integrations add inherited, inconsistently documented state and lower repetition. An established SaaS procurement catalog could make Buy earlier than Build, and a highly standardized acquisition program could reverse the final ordering. A calendar assessment should report, for each gate, the observed horizon on its actual tasks, its error and labor components, qualification status, and effective mandate.

Updating a model name without those quantities is not evidence that a milestone has been reached.

16 The objections that could break the argument

16.1 Reliability and distribution shift

An 80% task horizon is not 99.9% gate reliability. Even zero failures in a trial cannot establish universal safety. With n independent, representative Bernoulli trials and no failures, the one-sided 95% binomial upper bound is $1 - 0.05^{1/n}$. Reaching 10^{-3} needs 2,995 such trials; reaching 5×10^{-5} needs 59,914. Repeated prompts on the same few dossiers do not create that many independent cases. Volume is the practical difficulty: a route with five visits a month yields sixty visits a year, and sixty visits without failure bound the error rate only below 4.9%. A year without incident on a low-volume route therefore cannot support a 0.1% ceiling. The qualification plan must say where the volume comes from, off-production dossiers, dedicated campaigns, or several sites, and why those cases are representative and independent; sharing a service across customers helps to accumulate observations without making their contexts identical. Changes in policy, supplier population, or tools can invalidate transport of the estimate.

A defensible system is qualified on declared domains and monitored for drift; unknown cases trigger a valid recovery, inquiry, or refusal. If the accepted contract requires a human recovery, the replacement definition has not been met. If the system merely refuses an expanding fraction of legitimate work, its false-refusal and utility metrics fail. Neither a high average score nor an unlimited abstention policy closes a gate.

Reliability is also a path property. Let $\mathcal{C}_i$ be correctness of gate i under an independent reference standard. The probability of no gate errors is exactly

$$\Pr\left(\bigcap_{i=1}^n \mathcal{C}_i\right) = \Pr(\mathcal{C}_1) \prod_{i=2}^n \Pr\left(\mathcal{C}_i \mid \bigcap_{j<i} \mathcal{C}_j\right), \quad (26)$$

when the conditioning events have positive probability. Twenty marginal accuracies of 0.99 imply a no-error probability of $0.99^{20} \simeq 0.818$ under independence, but only the Fréchet bounds 0.80–0.99 without a dependence model. Two agents using the same model, policy interpretation, or false source may agree and fail together. Common failures can have worse severity despite a higher probability of an entirely correct run. Route tests must measure dependence and loss, not infer diversity from the number of agent instances. Published evaluations cover pieces of this problem: enterprise web tasks (Drouin et al., 2024), operational IT scenarios (Jha et al., 2025), long software-task completion (Kwa et al., 2025), and contextual integrity in enterprise workflows (Fu et al., 2026). They motivate separate utility, reliability, and information-flow evaluation; they do not supply the DGF scenario parameters, and they establish present limits, not permanent ones.

16.2 The adversary and the project seeking approval

A supplier document is also an attack surface. AgentDojo evaluates how untrusted tool-returned data can redirect agents toward malicious actions (Debenedetti et al., 2024). A DGF agent reads contracts, tickets, and inventories supplied by parties with interests in its disposition. Instruction filtering alone is not an assurance argument; external text must not gain the authority to change rules or grant tools.

Projects can also optimize the measured criteria without any injection. The allocation of effort across differently rewarded activities connects to multitask incentives (Holmström and Milgrom, 1991); the distinction between improvement and gaming is explicit in Kleinberg and Raghavan

(2019); strategic classification supplies the broader setting in which evaluated people change the features presented to a decision rule (Hardt et al., 2016). Consider a minimal gate model. A project begins with true compliance ξ . It chooses substantive remediation $u \geq 0$ and cosmetic reporting effort $g \geq 0$, producing observable score $y = \xi + u + g + \varepsilon$. A threshold τ determines preliminary approval. Let V be the project value of that approval, $c_u(u)$ and $c_g(g)$ the two effort costs, and $D(g, a)$ the expected sanction, lost approval value, and remediation cost from detecting misleading reporting under assurance intensity a . The project chooses

$$\max_{u, g \geq 0} V \Pr(y \geq \tau) - c_u(u) - c_g(g) - D(g, a). \quad (27)$$

Let $p(s) = \Pr(s + \varepsilon \geq \tau)$ and $s = \xi + u + g$. At a differentiable interior optimum, $Vp'(s) = c'_u(u) = c'_g(g) + D_g(g, a)$. Assurance raises the marginal cost of gaming when $D_{ga} > 0$. With $\omega_u = c''_u > 0$, $\omega_g = c''_g + D_{gg} > 0$, $t = Vp''(s) \leq 0$, and $\Delta = \omega_u \omega_g - t(\omega_u + \omega_g) > 0$, implicit differentiation gives

$$\frac{\partial u^*}{\partial a} = -\frac{tD_{ga}}{\Delta} \geq 0, \quad \frac{\partial g^*}{\partial a} = \frac{(t - \omega_u)D_{ga}}{\Delta} < 0, \quad \frac{\partial s^*}{\partial a} = -\frac{\omega_u D_{ga}}{\Delta} < 0. \quad (28)$$

Stronger assurance reduces cosmetic effort and, with diminishing marginal approval returns, weakly increases genuine remediation. The observable score still falls, so a lower pass rate after assurance is strengthened is not by itself evidence of worse projects. For smooth zero-centered symmetric noise with density f strictly decreasing away from its mode, $p''(s) = -f'(\tau - s)$; hence $t \leq 0$ when $s \geq \tau$, where preliminary passage probability is at least one half. Below the threshold, where $t > 0$, both efforts can fall. This predicts heterogeneous responses by pre-intervention distance to the threshold. Flat density segments and corner solutions require separate treatment; clearer criteria can lower both effort costs, leaving their allocation effect ambiguous. The ancillary mathematical notes give the derivation and boundary checks.

The model identifies a testable failure mode: declining escalations can coexist with concealed defects. A falling exception rate can reflect improved systems, better extraction, or concealed exceptions. Readiness inferred only from submitted documentation becomes less informative as projects learn the gate. An enterprise test compares claimed and independently verified compliance, uses hidden holdout checks of stable requirements, and records post-approval defects. Provenance checks, random audits, independent validation, and restricted authority all enter the assurance cost. The law requires automation that remains effective against adaptation, not a faster route to approving misleading evidence.

16.3 Who verifies the automated verifier?

The assurance argument is a division of mechanisms, not an infinite stack of identical language-model opinions. Typed outputs and capability-limited workers enforce action constraints. Independent retrieval checks provenance against authoritative systems. Deterministic tests establish narrow invariants. Differently constructed verifiers and held-out adversarial tests address correlated interpretation errors. Recorded outcomes expose deterioration over time. Model diversity only helps when the relevant errors actually differ; a shared false policy can defeat every model. An agent must not be able to create its own policy exception, alter the evidence against which its output is tested, or confer on itself a privileged tool.

A safety case then states the accepted residual risk, domains, fallback, revocation conditions, evidence of performance, and accountable operator. Vendor responsibility, contractual remedies, and insurance can allocate loss and create incentives; they do not establish correctness or independence. A mandatory human audit or intervention remains human work in B_A or h_E . Bainbridge’s analysis explains why supervisory and recovery costs cannot be ignored and argues

for continuing human involvement (Bainbridge, 1983, pp. 775–777). The law of gate substitution challenges the permanence of that residual, not the existence of those costs.

The strong thesis predicts that this recurring assurance work itself becomes executable under accepted rules. It can fail: new policies or distribution shifts may continually require human judgments that no validated procedure replaces. A standing mandate solves the authority of execution only within its scope; it does not solve the epistemology of assurance or discharge statutory management duties.

16.4 Why not repeat profession-wide automation predictions?

A diagnostic classification is not a whole radiology practice, and vehicle control is not every function in road transport. The same warning applies here: passing a checklist is not the whole DGF. The advantage of the gate unit is that it enumerates the inquiry, output, authority, and support that remain before claiming replacement. It also provides a finite case registry against which failure can be observed.

The informational character of the work is an advantage, not a metaphysical guarantee. Many DGF inputs and actions are already digital; others require new observations, negotiation, or physical inspection. A human collecting indispensable evidence outside the buyer’s boundary still performs necessary work. Appendix C makes the information limitation precise. The Integrate route is included because inherited evidence and low repetition test this weakness rather than allowing the thesis to be demonstrated only on a clean software pipeline.

16.5 A difficult residual is not a permanently human residual

Knowledge hierarchies concentrate expert work on problems that other workers cannot resolve (Garicano and Rossi-Hansberg, 2015, Sections 1 and 3). Automation can initially reproduce that arrangement: machines handle standard cases and people absorb exceptions. The law does not confuse this intermediate allocation with its endpoint. The exception handler has its own observations, hypotheses, tests, and outputs; it belongs to the substitution target. An undocumented acquisition incident may require fresh evidence, not better text generation. Supplier negotiations require interaction with another party. A new project’s failure can reveal an unknown architecture rather than a missing checklist field. The fact that a current system cannot close these activities is evidence about the remaining technical work, not a proof that the role must always be human. Hypothesis H6 turns this objection into a measurement.

16.6 Exhaustion or renewal: the rival dynamics

Continual automation is not enough to reach zero. Let R_t be the residual human load, a the fraction of it closed in each period, and b the new human needs that appear in each period, some of them created by the automation itself:

$$R_{t+1} = (1 - a)R_t + b, \quad 0 < a < 1, \quad R_\infty = \frac{b}{a}. \quad (29)$$

With constant $b > 0$, automation keeps progressing and the human requirement never disappears. The law therefore needs more than $a > 0$: it needs $b_t \rightarrow 0$, new indispensable human work that stops appearing. Convergence is also weaker than closure: a load that tends to zero can still be positive at any finite date, as Section 12 notes for posts. Such a residual is compatible with the separate 80% FTE reduction predicted for 2033.

The reason to expect exhaustion rather than renewal is that the work created by automation has the same shape as the work it replaces. Supervising, evaluating, triaging exceptions, and maintaining agents all read records and return artifacts, so the same substitution applies to

them, one level up. If each unit of automated work creates $s < 1$ units of supervisory work and each level of supervision can in turn be executed under mandate, the human need that remains after k levels is of order s^k , not a constant. The rival is a level at which $s \geq 1$, or at which no mandate is accepted, for instance a statutory duty of the management body. This is an argument, not evidence, and it is an argument about contraction, not about closure: it explains why the residual can become small. Closure is the architectural condition of Section 4.1, an implementation in which no necessary human step remains, including for its maintenance, and it is established contract by contract, not as the limit of a sequence. Hypothesis H6 and the intervention registry of Section 17 are how the contraction is checked.

16.7 The economic rival: a small but valuable human residual

The strongest rival is not that inputs and outputs are absent. Gans and Goldfarb (2026) emphasize O-ring task complementarity: improving automated tasks can increase the importance of remaining manual bottlenecks and change the incentive to automate them. This gives a serious rival endpoint: a small but valuable human residual may remain economically preferable. The law of gate substitution needs the quality-adjusted machine frontier eventually to overtake that residual too. It cannot derive that crossing from the current size of a department. Section 21 places this rival among the closest economic models.

17 A test that can fail within a reader’s lifetime

This section specifies how the broader workforce hypotheses could be tested. It is a proposed field-research framework, not a registered or active enterprise study. The current experimental programme evaluates LLM decision capability on governance reviews. No human-comparison study, independent human labeling, recruitment, or enterprise enrollment is planned in that programme.

17.1 A dated FTE claim with a stable accounting boundary

The primary dated hypothesis is **80% fewer workload-equivalent FTE required by the DGF ecosystem by 20 September 2033 than in 2026**, for comparable governed output, quality, and service. Let $\bar{N}_y$ be mean monthly required FTE over the year ending on 20 September of year y . As in Equation (14), each monthly value is total human work L divided by a fixed useful-hours convention K . For $\bar{N}_{2026} > 0$, the hypothesis is

$$\bar{N}_{2033}/\bar{N}_{2026} \leq 0.20. \quad (30)$$

The synthetic example uses $K = 120$ hours per month and a baseline of 140 FTE, giving a 28-FTE threshold. An empirical study prespecifies and holds its conversion convention constant across years; changing working-time conventions cannot manufacture a reduction. Required workload-equivalent FTE, allocated staffing FTE, and individual headcount are reported separately.

The final observation window runs from 20 September 2032 to 20 September 2033; baseline records cover the corresponding twelve months ending in 2026. The protocol is proposed: no enterprise cohort has been enrolled and no preregistration identifier is claimed. As of this revision on 24 September 2026, the baseline window has already closed. The original prediction therefore requires an auditable retrospective reconstruction of that year; it cannot now be presented as a fully prospective cohort test. Missing baseline records prevent confirmation of this dated comparison. A later registered baseline with a seven-year follow-up would test a separate prospective hypothesis, as specified in the companion manuscript, and cannot move or confirm the original 2033 deadline.

The accounting boundary includes internal labor, contractors, external reviewers, supplier operators, automation supervision, maintenance, rework, recovery, and all new DGF support. Record active hours with disjoint activity codes; shared support is counted once before allocation across customers. A half-time contribution is half an FTE under the declared conversion, not one whole person. Outsourcing or renaming a team does not remove its work. Unobserved supplier effort is missing data, not zero FTE. The illustrative 140-FTE enterprise pool is not itself a measured ecosystem baseline.

Register organizations, activities, continuity rules, and supplier coverage before evaluation. New support for the registered DGF functions stays in scope. Trace mergers, exits, and scope changes; lost coverage cannot count as an FTE saving. Ecosystem-wide inference requires justified sampling and weighting; a convenience cohort can only test its covered population.

Measure both actual total FTE and FTE standardized to the baseline volume and case mix of governed outputs. Prespecify the matching or weighting method and keep fixed support as a separate term; it must not be scaled away when workloads change. Report new DGF activities and their support explicitly, with a sensitivity analysis for scope changes. A fall caused only by fewer projects does not confirm the primary comparable-output prediction. Quality and service must meet prespecified noninferiority margins, with valid authority. Attribution to AI requires a comparative design that addresses demand, restructuring, and other changes; an FTE ratio alone establishes no causal effect.

Complete execution and support closure remain stronger, undated outcomes. An unresolved human task can coexist with the 80% FTE reduction. Each registered contract still specifies evidence, dispositions, quality, authority, and the full human-work boundary; Appendix E supplies an illustrative contract.

17.2 Measurement before deployment and after delegation

The first study can use historical logs across all nine populations. Apply a versioned shadow-routing rule using only information that would have been available at routing time. Keep later effort measurements hidden from the rule to avoid selecting expensive cases by construction. With J the assigned residual and T baseline effort, estimate q , κ , and $\phi = \mathbb{E}[TJ]/\mathbb{E}[T]$, with uncertainty and stratification by route. This identifies where human work concentrates without deploying a single agent; it does not measure η , automation quality, or workforce replacement.

An optional field study, outside the current programme, would compare human, assisted, and no-human execution on adjudicated cases, including valid refusals, missing evidence, and attempted manipulation. A no-human run counts as replacement only when quality, authorized action, and downstream acceptance are met. Research evaluators establish quality independently; continuing evaluator work required for operation counts in the support ledger. Ordinary production time, exceptional investigation, after-hours repair, supplier labor, external quality review, and fixed maintenance have disjoint activity codes. Delays are recorded separately from active hours.

17.3 Producer-by-format experiment

The handoff experiment crosses upstream producer with artifact format (Table 18), stratified by route. It assigns eligible cases before production and measures upstream and downstream human effort, including rejected outputs. Downstream tools and substantive policies are common within each route. Related integration work packages are clustered by integration program to account for shared evidence and reviewers.

Two estimands are distinct. A *format-only* experiment fixes adjudicated facts and randomizes their presentation, measuring the effect of representation. A *production* experiment assigns pro-

Table 18: Four conditions separating production from presentation.

	Free-form artifact	Structured, evidence-bearing artifact
Human production	Human narrative/dossier	Human-authored typed record
Agent-assisted production	Agent narrative/dossier	Agent-authored typed record

ducer and format before facts are extracted, retaining omissions, inconsistencies, and verification failures as outcomes. Restricting analysis to correct agent outputs would select away part of the treatment cost. For case c in cluster j , estimate separate outcomes $v \in \{h, c\}$ (hours or cost):

$$Y_{cj}^{(v)} = \beta_0^{(v)} + \beta_A^{(v)} A_{cj} + \beta_F^{(v)} F_{cj} + \beta_{AF}^{(v)} A_{cj} F_{cj} + \gamma^{(v)\top} x_{cj} + u_j + e_{cj}. \quad (31)$$

Here A denotes agent-assisted production and F structured format. For total human-hours, H2a predicts $\beta_F^{(h)} < 0$; the format effect among agents is $\beta_F^{(h)} + \beta_{AF}^{(h)}$. For upstream production-and-verification cost, H2b predicts $\beta_A^{(c)} + \beta_{AF}^{(c)} < 0$, including tools and allocated assurance. Downstream quality, labor, and rework meet prespecified equivalence margins across structured conditions. All assigned cases remain in the analysis; coefficients are outcome-specific. This prevents simple standardization from being attributed to agent intelligence.

Shared teams, caches, and process learning can create spillovers. Cluster or time-block randomization can address them, with carryover specified in the analysis. Human time is measured directly or checked against sampled observation; entry-to-exit elapsed time includes waiting and is not an effort measure. Process mining helps reconstruct visits and routes (van der Aalst, 2016). Failed, abandoned, and censored cases remain in the reporting boundary.

17.4 Prospective tests of the six hypotheses

For H1, two independent assessors score readiness before agent outcomes are available. Prediction is evaluated on held-out cases or units against simpler duration, completeness, and model-configuration baselines. Assessor agreement and probability calibration are reported separately.

For H3, the shadow-routing study above identifies κ , ϕ , and their uncertainty within each route; substantive exceptions are separated from random audits or fixed authorization rules. Post-deployment effort on that partition identifies η . The frozen partition distinguishes baseline difficulty from additional recovery work caused by deployment. Variance and post-treatment duration are secondary outcomes, not automatic implications of selection.

For H4, a training period estimates $q, \kappa, \eta, \mu, r, B$ and visit rates; predictions are frozen for held-out cases or units. The uniform-effort comparator predicts $\hat{L}_U = \hat{\lambda} \hat{h}_0 [\hat{q} + (1 - \hat{q}) \hat{m}]$ with $\hat{m} = \hat{h}_R / \hat{h}_0$, setting conditional exception effort to the overall baseline and additional overhead to zero. Both models use the same training period. Total hours include maintenance and shared assurance; a stratified case-mix forecast supplies a second benchmark. Added parameters must improve held-out loss and direction-of-change accuracy, not merely training fit. The discriminating test then changes the routing rule, the case mix, or the agent configuration between training and evaluation: the component model re-estimates ϕ on historical work under the new conditions and carries η and μ over, while the matched comparator carries over its conditional means h_E and h_R . Ratios are estimated within sufficiently homogeneous categories of cases, and the compared models receive the same information and the same recalibration budget. A change in η between a historical period and a post-deployment period is attributed to the agent only when other differences, in population, policy, or practice, are controlled; a frozen routing rule helps but does not remove them all.

For H5, four records distinguish who produced analysis, executed tools, authorized the disposition, and held answerability for it. Time-series adoption slopes or time-to-delegation contrasts are evaluated where institutional arrangements allow variation. Immutable human-signature requirements are recorded as fixed strata rather than counted as evidence for a discovered lag.

For H6, every residual human intervention is logged with a category and an origin: present at baseline, displaced from another role or organization, or created by the automation (supervision, evaluation, exception triage, agent maintenance). Each annual evaluation reports, per route, the number of open categories, the hours they absorb, the categories closed, and the categories created. The taxonomy of categories is versioned: a category is closed only when a qualified capability executes it under mandate on the declared case population, not when it is merely absent from the cases observed, and renamed, merged, split, or displaced categories are traced so that the count cannot change without a change in the work. Reopened categories are reported. In the terms of Equation (29), the registry estimates the closure rate a and the creation rate b . Complete substitution predicts that closures exceed creations until the registry is empty; durable complementarity predicts a stable or renewing set. A registry can remain nonempty in 2033 while the FTE prediction succeeds. It then supports partial substitution but does not establish complete closure. This is the test that separates the two trajectories, and it is the one the law ultimately stands on.

A second, separate experiment compares no intervention, upstream only, downstream only, and both. It estimates the interaction in Equation (22). Positive factorial producer effects do not themselves establish that deploying adjacent gates together is superadditive.

17.5 Quality, adaptation, and the data boundary

A minimum case record contains pseudonymous case, parent program or project, route, and visit identifiers; population/activity codes; active hours; model/tool and policy versions; provenance; routing reasons; adjudicated outcomes; returns; upkeep; and authorization references. Independent assessors resolve contested cases under written criteria while retaining disagreement. Model self-reported confidence is not treated as calibrated correctness.

Before evaluation, the analysis plan specifies the smallest relevant labor saving, noninferiority margins for quality, forecast-error tolerance, clustering, sample-size assumptions, and stopping conditions for unsafe approvals or disclosures. Rare harmful events require uncertainty bounds and sufficient follow-up, not an inference of safety from zero observations. The strategic extension adds independently verified compliance and a pre/post-policy analysis so that apparent progress cannot be explained only by improved presentation of unchanged defects.

17.6 Deadlines, plateaus, and the last human task

Each annual evaluation reports the four terms of Equation (11), human minutes per visit, separate support hours, adjudicated quality, and mandate status. A support cost divided among more customers is reported separately from an actual reduction in support hours. At each route, the blocking term is named: evidence pursuit, judgment, verification, per-case approval, maintenance, or legal oversight.

A proposed companion trajectory test, to be preregistered, defines a plateau as a residual ratio above 5%, with less than 0.5 percentage points of reduction per year over three successive years, measured at comparable output, quality, and evaluation resources. The thresholds are design choices, not natural constants. Such a result rejects the continued-decline specification; it is warning evidence, not a logical refutation of the 80% FTE prediction. A plateau below 20% of baseline required FTE could satisfy that prediction while contradicting continued movement toward zero.

At the deadline, a comparable-output FTE ratio above 0.20 rejects the dated prediction for the observed population. Confirmation also requires the registered quality, service, and authority conditions. Report uncertainty in time measurement and standardization, with organizational clustering and shared-supplier dependence. An interval spanning 0.20 leaves the threshold result inconclusive; missing baseline or supplier effort prevents confirmation. Publish raw and standardized FTE together, and distinguish the threshold result from evidence of AI causation. The twelve-month window captures periodic review and maintenance; it does not establish zero risk on arbitrary future inputs.

17.7 What would defeat the law?

An established indispensable human intervention defeats technical universality for its declared domain. A legally required personal approval prevents complete adoption there while it remains necessary. A provider-operated hidden review queue defeats an apparent no-human API, and recurring human support defeats support closure. These findings need not reject an 80% FTE reduction: the claims have different thresholds.

The dated FTE prediction fails if the correctly measured ratio remains above 0.20 in 2033. It cannot be rescued by dropping difficult sites, excluding new support jobs, changing the FTE conversion or workload comparison, or postponing the deadline. A lower required FTE count can confirm the comparable-output hypothesis without a matching fall in individual headcount. A staffing cut alone cannot establish lower required work, an AI effect, or preserved governance quality.

Publish workload-equivalent FTE, allocated staffing FTE, individual headcount, qualified execution, effective mandates, and support closure separately. The unrestricted statement “one day, all gates” has no ordinary finite-date disproof; the 2033 FTE hypothesis supplies a refutable intermediate prediction without claiming that its success would prove the universal endpoint.

18 DGF-Bench: measured review performance on 300 synthetic projects

18.1 Question, population, and information available to the agents

This experiment asks whether an AI agent can apply a declared governance policy, identify the required findings and actions, cite observed evidence, respect its authority, and carry a review through a connected route. It measures review decisions and proposed actions, not implementation of remediation or changes in human workload. It is distinct from the prospective enterprise and factorial studies in Section 17.

Three practical questions organize the analysis: (1) does the model reach the disposition required by the rules, (2) does it produce all the supporting work needed for a valid review, and (3) does that reliability survive a sequence of reviews at an acceptable API cost? The unit called a *case* or *dossier* is one generated project; a *gate* is one checkpoint on its route. Three models reviewing the same 300 projects produces 900 planned model–case runs, not 900 independent business projects.

The dataset contains 300 fictional projects: 100 Buy, 100 Integrate, and 100 Build cases. Buy and Integrate contain six gate occurrences each, and Build contains five, for 1,700 occurrences per model and 5,100 planned observations across three models. Cases were generated before these evaluations at difficulty setting 4, with sampling conditioned to improve decision coverage. The setting controls candidate fault patterns, including address overlap, supplier sanctions, and mandatory-criterion failures; it is not an empirically calibrated measure of task difficulty. Facts

are rejection-sampled to balance base General dispositions and approximately balance feasible specialist dispositions while preserving route consistency. This conditioning changes the raw generator frequencies; no comparison across difficulty settings is reported. Project context, owners, budgets, data classifications, technical architectures, documents, and simulated system records derive from canonical case facts. Documents can be missing, stale, or inconsistent. Specialist reviews precede a final General review; the routes are examples rather than a universal enterprise standard.

The agent receives gate objectives, a structured output contract, an explicit decision policy, and tools to read allowed evidence and query the synthetic environment. In particular, the policy includes executable rule definitions and identifies an authoritative structured `REVIEW_FACTS` snapshot. The evaluator-only reference file is inaccessible through the agent’s tools. This is nevertheless a strongly scaffolded test of policy application: the agent need not discover the rules or reconstruct every decision-relevant fact from unstructured documents. Architecture images are available under the vision setting; image availability is not evidence that image interpretation was necessary for success. These design choices limit claims about expert reasoning on independently collected enterprise dossiers.

The dataset audit reports 300 distinct project identifiers, architecture signatures, and complete decision-fact signatures, but only 267 distinct whole-case finding patterns. Gate-specific relevant fact signatures include 142 of 200 Compliance cases and 241 of 300 Security cases. All rules applicable to the selected phases are covered according to the generator’s audit. Phase-adjusted normalized decision entropy ranges from 0.9971 to 1.0000. Balanced labels improve coverage, but do not estimate enterprise prevalence or establish independent business validity.

18.2 What is generated, and what the model actually does

The case generator is a program using seeded facts, rules, and document templates. The AI model under evaluation is the reviewer, not the author of the fictional need or architecture. A Buy scenario represents acquiring a solution; Integrate represents connecting existing systems; Build represents developing a new project. Those scenarios are given budgets, users, owners, dates, data classifications, and technical constraints. A generated name is a scenario label, not an independently elicited customer requirement.

1. **Construct the project facts.** The program fixes what is true in the synthetic case, such as the requested budget, whether a supplier has supplied a document, or whether a restore test succeeded. Decision-coverage sampling produces cases with different required outcomes.
2. **Generate the design and dossier.** The same case drives an architecture diagram, project charter, gate review requests, and evidence files. A reviewer may receive a technical design, network/data-flow tables, contract records, security evidence, or readiness tests, depending on the gate (Table 19). Some presented evidence is incomplete or stale.
3. **Start a gate review.** The harness supplies the project context, gate and phase, applicable policy, permitted evidence and tools, and the answer schema. The model decides which allowed records to inspect. It can list and read evidence, query simulated systems, request information, and inspect existing authorization mandates.
4. **Submit a decision.** The model returns a disposition, finding identifiers, proposed actions, evidence references and supporting excerpts, an authorization indicator, and a rationale. A typed submission is recorded. Producing a syntactically valid record alone does not satisfy the substantive score.
5. **Continue the route.** In the tested `agent` mode, the harness passes actual upstream model

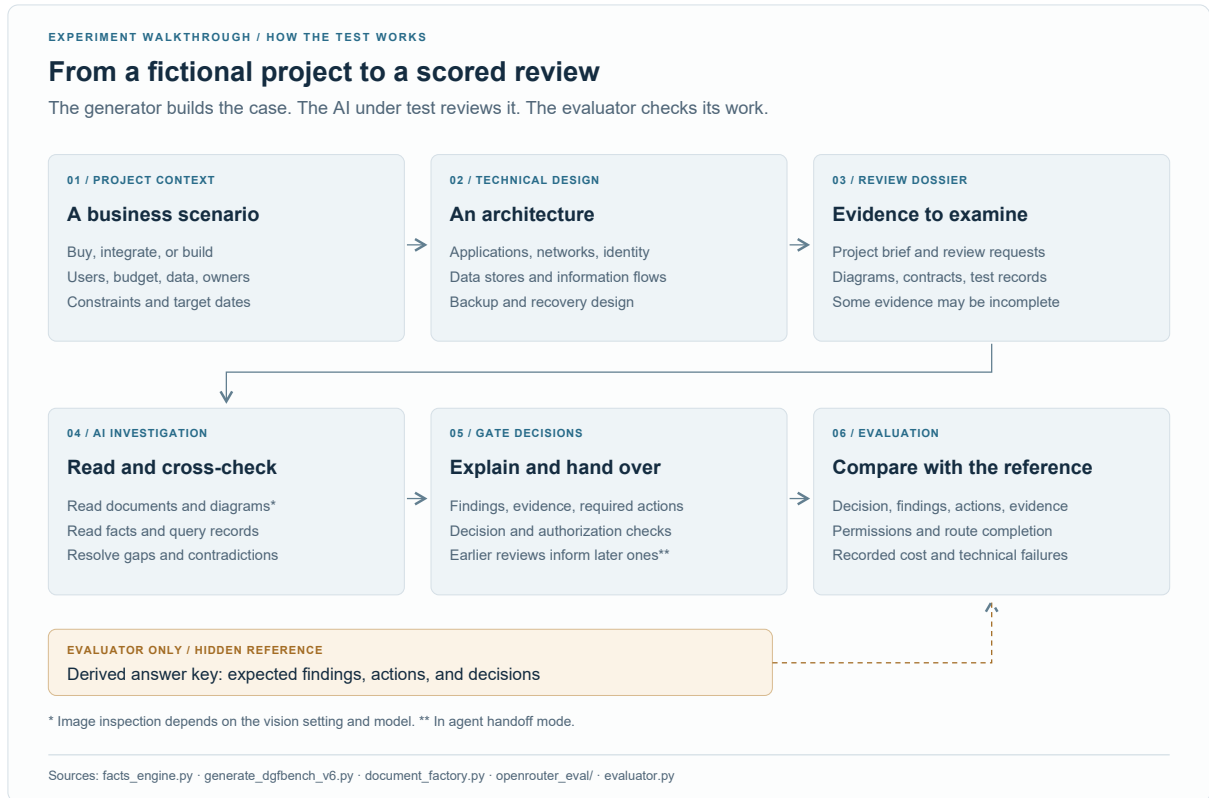


Figure 5: Experimental workflow. The generator constructs the project and evidence; the tested model investigates and submits reviews; the evaluator uses a separate reference. The diagram describes the implemented test, not a measured enterprise deployment.

reviews to later checkpoints. The final General review consolidates specialist opinions and its own checks. The tested routes are listed in Table 20.

- Evaluate and retain the trace.** A deterministic evaluator compares the submitted record and observed tool events with the reference. It records component scores, full gate and route success, critical misses, false approvals, provider failures, and usage. It is not another LLM judge. Failed or incorrect decisions are retained; infrastructure exclusions are reported.

Table 19: Concrete examples of generated review material. The contents vary with the route and gate. Authoritative structured snapshots are available in addition to these artifacts.

Material	Examples in the case	Review question
Project context	Charter and review request (Word), project context (JSON)	What is proposed, for whom, at what cost, and under whose ownership?
Architecture	Diagram (SVG/PNG), technical design, network and flow tables	Do the design facts meet the declared constraints?
Security	Identity, vulnerability, monitoring, and resource records	Are the protections required by the policy present?
Supplier and legal	Vendor evidence, contract versions, data-processing agreement status	Are the supplier and contractual prerequisites satisfied?
Readiness	Backup, restore/failover and load-test records; runbook	Can the proposal demonstrate the required operational readiness?

The five dispositions are **GO** (proceed), **GO_WITH_RESERVATIONS** (proceed under conditions), **REWORK** (revise the proposal), **SUSPENSION** (wait for a prerequisite), and **NO_GO** (do not proceed).

Table 20: The three complete route sequences evaluated in this study; each project ends with General.

Route	Cases	Ordered gates
Buy	100	Procurement, Legal, Compliance, Security, IT, General
Integrate	100	IT, Architecture, Security, Legal, Compliance, General
Build	100	IT, Architecture, Security, Tech Readiness, General

as submitted). Approval is therefore not the target outcome for every dossier: a justified refusal is a successful review. Tool actions take place in the synthetic environment; a request to fix a defect is not evidence that the defect was fixed.

18.3 A generated project and a recorded successful review

Case DGF-BLD-035200_build, *Project Falcon—Secure Vendor Portal*, provides a concrete example. Its generated HR context has 1,000 users, a requested budget of EUR 500,000, confidential data, high business criticality, and a primary region of UAE North. Figure 6 reproduces its actual evidence diagram. It depicts identity, application and data services, monitoring and backup, and a secondary recovery region. Its labels are claims in the dossier to be checked, not an independently validated production blueprint.

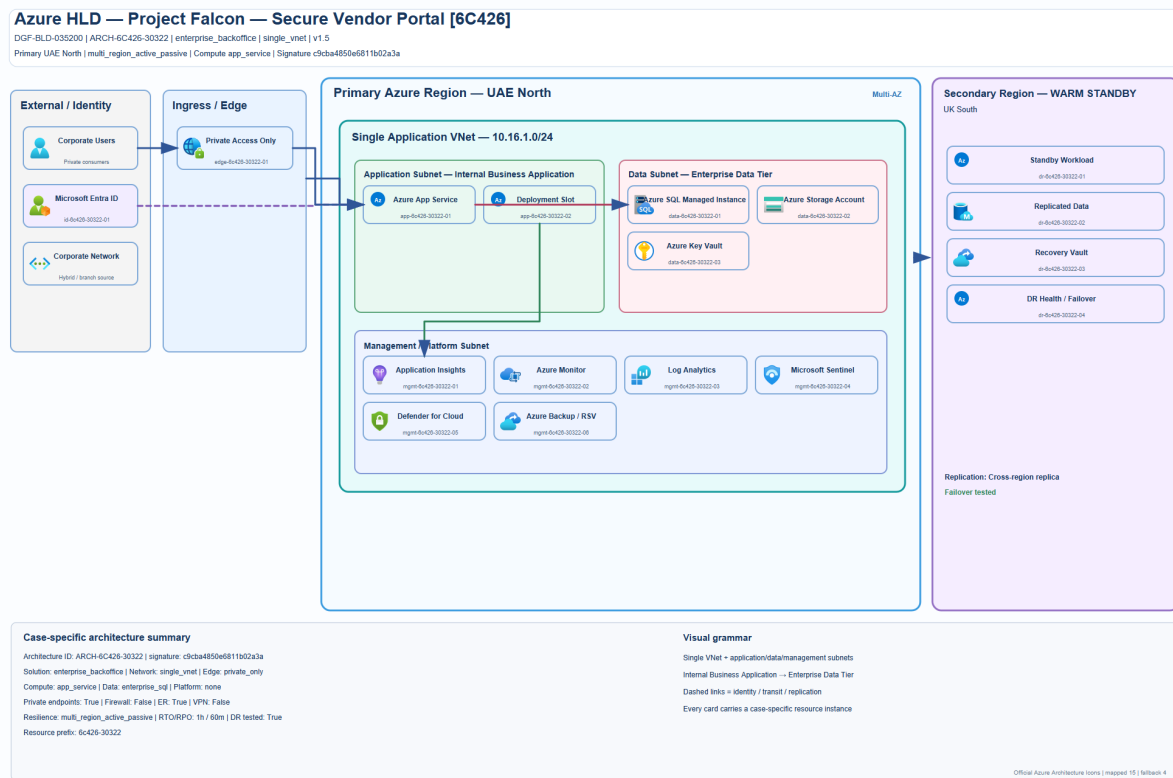


Figure 6: An actual generated architecture document from Project Falcon. Full-resolution evidence is in the released dataset. The figure illustrates what a dossier can contain; it does not imply that this image was necessary to obtain the correct review outcome.

In Gemini’s recorded Tech Readiness review, the authoritative record has a draft runbook. The agent identifies TR-OPS-001 and proposes completing the operational handover. Other required readiness checks pass. It cites the observed excerpt "**runbook_status**": "**draft**", consults the active mandate, and executes the simulated conditional-approval tool with a condition to approve the runbook before production cutover. The tool reports **EXECUTED** and explicitly leaves

the finding unresolved. The submitted outcome is therefore `GO_WITH_RESERVATIONS`, with the finding and required action retained and the authorization indicator set to true.

All three models pass all five gates of this particular dossier. Their observed dispositions are `GO`, `GO`, `GO`, `GO_WITH_RESERVATIONS`, and `GO` in route order. This example was selected after evaluation to explain the mechanics, not as a representative sample or an additional experiment. The study-level counts include every evaluable case, including unsuccessful reviews. The public research directory includes `worked_examples.json`, with the original archive-relative paths and selected recorded outputs for this example and the evidence-failure example below. The complete traces remain in the release.

18.4 Protocol, completion, and outcomes

Table 21 identifies the three tested endpoints. Each receives the same case population, policy, tool interface, and task limits; no model is trained or fine-tuned by this experiment. The comparison concerns these deployed endpoints under the recorded conditions, not different implementations of a human review process. Configuration was temperature 0, vision `auto`, agent-produced handoffs, at most 20 turns and 40 tool calls per gate, and 8,192 output tokens per turn. Gates remained sequential within a project. The last resume used 12 workers, four per model, and a USD 110 cost cap; an earlier invocation used six workers and a USD 65 cap. Data collection spanned 22–23 September 2026. Resumption retained completed cases and valid gate checkpoints. Historical provider/key-limit failures remain in the released traces. The 78.73-minute elapsed time in the last run statistics describes that resume only.

Of 900 planned model–case runs, 899 are evaluable: 300 for DeepSeek, 299 for Gemini, and 300 for Luna. Gemini case `DGF-INT-035128_integrate` failed at its first IT gate with a provider generation error (`finish_reason=error`); it is excluded as an infrastructure failure. Thus 5,094 of 5,100 planned gates are evaluable, with all expected gates attempted in the evaluable cases. There are no recorded final agent-protocol failures or model-identifier mismatches. Provider-reported identity is not an independent audit of underlying weights. DeepSeek was served by multiple OpenRouter providers; Gemini and Luna were recorded through Google AI Studio and OpenAI.

Table 21: Models used in the experiment. Names are short labels for the exact requested OpenRouter identifiers, not claims about independently verified model weights.

Model label	Requested identifier	Recorded serving provider
DeepSeek v4.1 Flash	<code>deepseek/deepseek-v4.1-flash</code>	Multiple providers
Gemini 3.8 Flash	<code>google/gemini-3.8-flash</code>	Google AI Studio
GPT-5.6 Luna	<code>openai/gpt-5.6-luna</code>	OpenAI

Scoring uses `DGF-decision-v8.2-authorized-review`. A gate succeeds strictly only if its disposition, finding set, action set, evidence provenance, and authorization indicator all satisfy the reference. Findings and actions require F1 equal to one. Evidence requires an observed reference and exact excerpts supporting the relevant field/value tokens. This is lexical provenance checking, not independent verification of arbitrary reasoning. Authorized conditional approvals can change the effective reference disposition; the evaluator verifies the mandate and recorded action. General combines effective reference opinions, so an upstream error can also cause a downstream failure. Complete-route success requires every gate to succeed strictly; it does not mean that remediation was executed.

For model m , let s_{mcg} be one when gate g of case c passes every component, and zero otherwise.

The two principal rates are

$$\text{GateCSR}_m = \frac{\sum_c \sum_{g \in G_c} s_{mcg}}{\sum_c |G_c|}, \quad \text{RouteSuccess}_m = \frac{\sum_c \prod_{g \in G_c} s_{mcg}}{|C_m|},$$

where C_m contains the evaluable cases and G_c their expected gates. This gate rate is weighted by gate count: a six-gate route contributes six observations and a five-gate route contributes five. A model passing four gates of a five-gate case has 80% gate success for that case but zero complete-route success. Decision-only accuracy checks just the submitted disposition. A critical miss is an omitted reference finding classified as critical; a false approval is `GO` or `GO_WITH_RESERVATIONS` when the effective reference requires rework, suspension, or refusal.

Gate confidence intervals use a route-stratified bootstrap of whole cases, preserving dependence between a project’s gates (2,000 draws, seed 81931; a case-count Wilson fallback handles degenerate intervals). Route intervals use Wilson bounds. Pairwise differences use the same 299 cases for all models, resampled together within routes (10,000 draws, NumPy seed 81931). These are descriptive intervals conditional on the synthetic sampling design. No preregistered analysis or independently adjudicated ground truth is claimed. The original 300-case study contains one trajectory per model and case; the subsequent repeated-trajectory study in Section 18.7 addresses variability on a separately identified 15-case subset.

18.5 Overall and route-level results

Table 22: Measured synthetic benchmark outcomes. Gate CSR requires complete conformity; route success requires all gates. Cost includes recorded failed attempts.

Model	Cases	Gate CSR [95% CI]	Route success [95% CI]	USD
DeepSeek	300	74.18% [71.88, 76.47]	24.67% [20.13, 29.84]	10.09
Gemini	299	94.98% [93.86, 96.04]	76.92% [71.82, 81.34]	71.88
Luna	300	83.29% [81.29, 85.24]	42.33% [36.87, 47.99]	5.05

Gemini succeeds strictly on 1,609 of 1,694 gates, Luna on 1,416 of 1,700, and DeepSeek on 1,261 of 1,700. Corresponding complete-route counts are 230 of 299, 127 of 300, and 74 of 300 (Table 22). Mean weighted component scores, which are not strict success rates, are 99.68%, 97.67%, and 95.02%, respectively. The distinction matters: strong average components can coexist with many incomplete project reviews.

On the 299 common cases, CSR is 94.98% for Gemini, 83.23% for Luna, and 74.09% for DeepSeek. Paired differences are 20.90 percentage points for Gemini minus DeepSeek (95% interval 18.30–23.44), 11.75 for Gemini minus Luna (9.62–13.93), and 9.15 for Luna minus DeepSeek (6.67–11.63). The ranking persists on the matched cohort; this is not a universal model ranking.

Table 23: CSR and complete-route counts by project route. Gemini excludes one Integrate case.

Route	DeepSeek: CSR; routes	Gemini: CSR; routes	Luna: CSR; routes
Buy	67.83%; 13/100	97.67%; 87/100	74.67%; 20/100
Integrate	78.17%; 32/100	95.12%; 75/99	87.67%; 52/100
Build	77.00%; 29/100	91.60%; 68/100	88.40%; 55/100

Gemini’s route success falls from 87 of 100 Buy cases to 68 of 100 Build cases. Luna succeeds on 20 Buy, 52 Integrate, and 55 Build cases; DeepSeek succeeds on 13, 32, and 29, respectively (Table 23). These differences do not isolate the causal effect of route length, because gate composition and case content also differ.

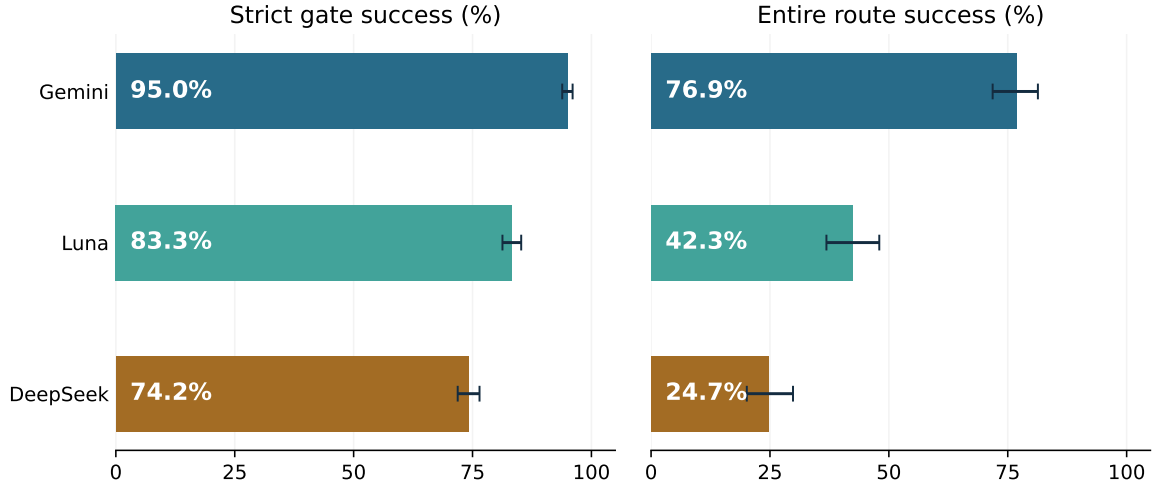


Figure 7: Measured gate and complete-route review outcomes. Intervals are computed over independent case units rather than treating all gates as independent trials. These are model observations on synthetic dossiers, separate from the workforce scenario figures.

Table 24: Strict success by gate family (percent). Counts vary by route composition; complete denominators and intervals are supplied in the released tables.

Gate	DeepSeek	Gemini	Luna
Architecture	75.00	85.43	89.50
Compliance	88.00	100.00	96.00
General	68.67	87.96	61.67
IT	87.00	100.00	97.67
Legal	68.50	99.50	80.50
Procurement	43.00	96.00	33.00
Security	72.33	98.66	92.00
Tech Readiness	71.00	89.00	97.00

Procurement is the lowest-CSR gate for both Luna (33%) and DeepSeek (43%). Luna’s General CSR is 61.67%. Luna exceeds Gemini’s descriptive CSR in Architecture and Tech Readiness (Table 24); these local comparisons are exploratory, without multiplicity-adjusted tests or a claim of specialist superiority.

18.6 Evidence failures, unsafe approvals, and cost

Disposition accuracy alone is 100% for Gemini, 98.12% for Luna, and 95.24% for DeepSeek. All 85 Gemini strict failures concern evidence conformity; findings, actions, dispositions, and authorization indicators match the reference. Evidence is the sole failing component in 221 of 284 Luna failures and 338 of 439 DeepSeek failures. Invalid excerpts and missing or unobserved references are grouped by these diagnostics; not all are harmless formatting defects. The post-hoc structural sensitivity and source audit below now separate some of these causes. Their alternative endpoint is explicitly labelled and does not replace the original strict score or claim to measure general semantic entailment.

A recorded example of a correct decision with a failed proof. In DGF-BLD-035201_build, Gemini reads the Architecture facts: an API gateway is required but absent, and measured latency is 55 ms against a 20 ms target. It correctly names ARCH-API-001 and ARCH-PERF-001, proposes adding the gateway and remediating latency, and submits REWORK. Two attempted

conditional approvals are rejected by the tool: one omits an open finding and the other includes a finding ineligible for risk acceptance. The submitted refusal to proceed is consistent with those restrictions.

The gateway excerpt reverses two fields. It places `api_gateway_present` before `api_gateway_required`; the observed record has the reverse order. The values are correct but the excerpt is not an exact observed sequence. Evidence fidelity is 0.5 for the two findings, so strict success is zero while disposition, finding, action, and authorization scores are all one. The weighted component score is 0.95. This selected trace explains exactly how a substantively correct review can fail this benchmark’s evidence contract. The original strict outcome is retained; the following sensitivity reports the effect of accepting exact field/value equivalence separately.

DeepSeek records one critical finding miss and one false approval, both in the Architecture review of `DGF-INT-035136_integrate`. An address overlap required `NO_GO`; at forced finalization, the model submitted `GO` with an empty finding set, no evidence, and a “placeholder” rationale. The response passed the submission schema but failed the substantive score. This observation combines decision quality with the ability to finalize under resource limits.

Luna records three critical misses and one false approval. In case `DGF-INT-035197_integrate`, its Legal review approved a dossier with a missing DPA, while its own rationale said that `REWORK` was required. The other critical misses concern private endpoints in Security and an upstream `NO_GO` in General. The latter can include error propagation and should not be interpreted as an independent new safety event. Gemini has no observed critical miss or false approval in its evaluable runs; this does not establish zero risk. Recorded truncated-response counts are 2 for DeepSeek, 14 for Gemini, and 0 for Luna, without implying that these counts equal incomplete cases.

The usage ledgers sum to USD 87.015829172, agreeing with the cumulative budget counter. Gemini accounts for USD 71.88, DeepSeek USD 10.09, and Luna USD 5.05, including recorded retries and failures. Approximate cost per evaluated dossier is USD 0.240, 0.034, and 0.017, respectively. Luna therefore combines a higher overall strict score with lower recorded cost than DeepSeek in this execution. Costs depend on provider routing, caching, price, and retry history; they are not a labor-cost comparison or a permanent pricing claim.

18.7 Completed controls, evidence audits, and repeated trajectories

The following additions distinguish completed analyses from development work and proposed experiments. They preserve every original score above. Detailed methods and machine-readable results accompany the focused companion manuscript, *DGF-Bench: Rule Application and Evidence Reliability in Synthetic Governance Reviews*. The companion analyzes the same original population and the completed repetitions; it is not an additional independent 300-project replication.

An executable-rule control. A deterministic comparator applies the policy already supplied to agents to the public factual snapshots. Generation blocks hidden-reference reads and is separated from scoring. It passes 1,700/1,700 gates and 300/300 routes, with no model API cost. It retains base decisions rather than requesting conditional approvals and may cite complete observed snapshots under the original unrestricted excerpt contract. This establishes that the scaffolded task is solvable without a language model. High LLM decision accuracy therefore demonstrates an ability to apply these rules through the interface; it does not establish added accuracy over the rule control or the ability to recover the necessary inputs from arbitrary documents.

Structural sensitivity on all three models. All 690 originally evidence-failed gates were audited. The alternative endpoint retains original lexical successes and additionally accepts exact field/value subsets within a single observed object, ignoring field order and equivalent numeric serialization. Required source reads, citations, relevance to a rule field, and all other score components remain unchanged; cross-object joins are not promoted. Table 25 shows the paired results. These are post-hoc provenance sensitivities, not a new semantic score: an observed value need not supply every premise of an argument. In particular, failure to parse an excerpt as one JSON object does not show that its factual claims are false.

Table 25: Original trajectories: strict (S) and post-hoc structural (R) success counts. Original primary outcomes remain unchanged.

Model	Gates		Complete routes	
	S	R	S	R
DeepSeek	1261/1700	1313/1700	74/300	85/300
Gemini	1609/1694	1678/1694	230/299	284/299
Luna	1416/1700	1454/1700	127/300	144/300

A separate Procurement audit examined 205 submitted support entries across 185 failed findings in 127 gates. DeepSeek supplied 25 exact rows from genuinely read and cited CSVs, covering 25 findings in 23 gates; each identifies the selected vendor. The snapshot-only source-ID requirement excludes these alternatives. Another CSV excerpt joins observed rows with ellipses. All 205 support entries cite observed sources. This supports distinguishing contract violations from fabricated evidence, while retaining incomplete quotations and unproven relationships as separate limitations. A pricing row alone, for example, does not establish the budget envelope.

Authority is exercised differently by each agent. On the 299 common cases, the same 391 non-General gate occurrences are eligible for conditional approval under the supplied policy and active mandate. Revalidated approvals are used on 391/391 by Gemini, 347/391 by Luna, and 201/391 by DeepSeek. Eligibility alone is not counted: the recorded action must validate and the final decision must use it. Separately, Gemini makes 466 rejected approval calls across its original retained checkpoints, sometimes repeatedly within a gate. Its absence of false final approvals therefore describes the model-plus-tool system, including enforcement by the environment. Conditions are checked as nonempty strings, not assessed for their business adequacy. Base-reference agreement and effective-reference agreement must likewise remain distinct; authorized actions legitimately change some targets.

Completed repetition study. Fifteen existing dossiers, five per route, were selected with a fixed seed before new outcomes. Three additional trajectories per model-case yield 135 evaluable runs and 765 gates; historical trajectories are not pooled as a fourth repeat. Table 26 reports strict and identically computed structural outcomes. Dispositions are correct on 245/255 DeepSeek, 255/255 Gemini, and 246/255 Luna gates. No false approval or critical miss occurs in these repetitions. All-three success concerns only 15 distinct dossiers per model, and the repeated gates are not independent observations. Two Gemini trajectories retain seven previously completed prefix gates after infrastructure resumption; those gates count once. There is no best-of-three selection.

Recorded repetition spending, including failed attempts, is USD 12.5598870228. Together with the unchanged original USD 87.015829172, the total is USD 99.5757161948. Route-stratified case-cluster bootstrap intervals retain all three trajectories of each sampled dossier; the small subset supports exploratory repeatability estimates rather than enterprise prevalence.

Table 26: Completed repetitions: S denotes strict success and R the same post-hoc structural endpoint as above. All-three counts dossiers whose entire route passes in each trajectory.

Model	Gates /255		Routes /45		All three /15	
	S	R	S	R	S	R
DeepSeek	187	195	11	11	0	0
Gemini	245	249	35	39	9	12
Luna	211	217	19	23	3	5

Input sufficiency and work not yet evaluated. A source inventory covers all 300 dossiers and 1,700 scheduled gates, with explicit decoders for 72 of 76 inventoried fields. Two counterfactual generator checks prove omissions from non-snapshot artifact contents: changing `it.duplicate_capability` or `compliance.audit_trail` changes the corresponding decision while all non-snapshot artifact payloads and rendering inputs remain identical. Uncertified fields are not automatically absent; phase access and alternative tools matter. In particular, `request_evidence` can return the full factual snapshot and must be controlled in any removal experiment.

Separate extensions add the two missing records to three new development cases, with access and counterfactual checks. A source-location citation prototype copies observed values by pointer or text span. Neither is an evaluated model improvement or a held-out benchmark. A targeted General-only study is prepared but unrun: 270 fresh reviews compare normalized agent histories with references adjusted to the same validated approvals. Its selected disagreement cases estimate a conditional handoff effect, with identical-history controls; no result or expense from this preparation enters the measurements above.

Reproduction records and linked item-level inventories are under `research/2026-09-followup/`. Its README links the rules-control summary, all-model evidence audit, approval events, Procurement sources, repetition sensitivities, and source-coverage inventory. The public release retains the source dossiers, traces, repetition archive, and follow-up artifacts.

18.8 Interpretation and reproducibility boundary

The empirical conclusions are concrete. **First**, Gemini has the highest overall strict review success under this protocol, including in the paired comparison. **Second**, a project-level requirement amplifies remaining strict-contract defects: 94.98% gate success leaves 69 of Gemini’s 299 evaluable projects without a completely conforming route; the separately labelled structural endpoint reduces that count to 15. **Third**, evidence conformity is the sole failing component in all Gemini strict failures and most failures of the other models. **Fourth**, Luna achieves higher overall strict success than DeepSeek at approximately half the recorded cost, while both retain substantive errors. Finally, the zero observed false approvals for Gemini are an observation about this sample, not a safety guarantee.

For an FDE, these results identify implementation work to investigate: preserving observed evidence through structured submissions, checking agreement between a rationale and its typed decision, handling turn-limit finalization, and inspecting downstream consequences. These are proposed engineering responses to observed failure modes. The completed diagnostics and repetitions above clarify those modes; the development prototypes have not yet demonstrated a model-performance gain.

The experiment demonstrates strong review performance under an explicit rule system: Gemini satisfies every strict requirement on 94.98% of evaluable gates and on every gate of 76.92% of evaluable project routes. This is empirical support for the technical prospect of substantial

automation of the review tasks represented here. The rule control also completes this fully specified task, so the observed LLM capability is not an accuracy advantage over deterministic software. If comparable performance transfers to operational settings, these tasks could require less human effort and fewer staff, including substitution for work currently performed by employees. That implication is conditional on the human work needed for exceptions, oversight, corrections, integration, and maintenance; a gate success percentage is not an FTE-saving percentage. The experiment supplies no direct estimate of review hours saved, autonomous remediation, production acceptance, human equivalence, or the magnitude and timing of the 2033 FTE hypothesis. The generator and evaluator share rules, facts are deliberately structured, decisions are balanced, and there is no human baseline or independent expert adjudication. Policy-application results should not be generalized to tacit rules, adversarial inputs, private enterprise populations, or complete governance automation.

The published release preserves all files of the recorded run and its dataset, including old failed attempts, the original partial exports, and the final analysis. Only the final analysis tables describe the 899 evaluable runs. Frozen benchmark sources reproduce the recorded source fingerprint; the repository’s evolving default branch is not substituted for them. Five targeted offline rescoring checks reproduced the saved gate scores exactly, including all cases with critical misses, and all 899 score files passed internal count consistency checks. These checks establish reproducibility of this evaluator, not independent validation of its governance judgments.

19 Implications if the hypotheses hold

19.1 Governance can expand while processing labor contracts

A gate does not have to disappear for its economics to change. If verified checks become cheaper, the enterprise can perform more of them at the same budget: more projects reviewed, more conditions traced to closure, and more frequent reassessment. The paradox is that a stricter DGF can require less routine case-processing labor. The gain depends on evidence quality and total assurance cost; the relevant choice is how to use released capacity, not whether every formal review is removed.

This implication is stronger than a claim that agents write reports faster. The capacity frontier changes the set of control programs the enterprise can afford. A firm can purchase broader coverage, faster decisions, or more expert attention to difficult cases. These are alternative uses of the same resource gain and cannot all be counted as independent financial savings.

19.2 Upstream interfaces can outweigh local headcount

The largest department need not be the most valuable first intervention. An acquisition’s inherited-system evidence or a new project’s architecture-and-permission record can affect several specialist populations simultaneously. Its value comes from the downstream work it changes, including avoided returns. Conversely, an upstream step with few consumers or expensive reconciliation may have little leverage.

If Hypothesis H2 holds and measured portfolio interactions are favorable, prioritization should follow verified dependency effects, not only local employee counts. The appropriate comparison is the full counterfactual in Equation (20). This is upstream leverage, not a rule that earlier is always better or that every cascade reinforces itself.

19.3 Departments become control planes

In the exception-routed configuration, some specialist effort moves from individual decisions to the system that produces decisions. Procurement maintains supplier-evidence contracts; Legal

maintains approved clause interpretations and escalation rules; architects maintain reusable patterns; Security maintains control tests and adversarial evaluations; IT production maintains recovery checks and execution limits; Compliance maintains the mapping of rules to controls. Humans still handle novel cases and own consequential authorizations where retained.

A *control plane* is this layer of policy, evaluation, monitoring, and delegated authority over the case-processing layer. Its work is precisely the B_A and shared assurance that a credible business case must count. The organizational shift is from applying expertise afresh to every case toward encoding, testing, and revising the conditions under which execution is delegated. Bainbridge’s skill-retention problem remains relevant: a small team cannot safely oversee rare failures without opportunities to maintain operational competence (Bainbridge, 1983).

19.4 Firm boundaries and gate redesign remain choices

Lower coordination costs, internal or external, can move the boundary of the firm in either direction (Section 21). Companies may also combine gates, remove duplicate reviews, or create a shared evidence service rather than automate the historical sequence. The DGF is an initial architecture to measure, not a requirement to preserve every inherited handoff. This is compatible with Law 3: what must be preserved is every obligation and the logical order of authorizations, not each inherited box. Inside those choices, the gate model identifies what must be measured: whose work is removed, whose work is added, what is verified, and where authority resides.

20 Who closes the road: forward deployed engineers

Like Section 19, this section states implications of the thesis: the author’s expectation, not a result. The law says what becomes software. It does not happen by itself. In practice the work is done by a particular kind of engineer. A forward deployed engineer (FDE) works alongside customer teams to scope, build, integrate, and deploy software in their operating environment. OpenAI describes an end-to-end role spanning discovery through production rollout; Palantir describes embedded engineers building customer-specific applications and owning implementation (OpenAI, 2026; Palantir Technologies, 2026). These role descriptions establish what the organizations ask FDEs to do, not that the role has already automated a DGF or delivered the FTE reduction hypothesized here.

An FDE meets the DGF twice. First as an obstacle: an agent project passes through the same IT, Architecture, Security, Readiness, and General gates as any other change, and the Build case of Section 9 is exactly such a project. Then as an object: the FDE is the person who turns a general agent platform into gate implementations. That work is concrete. It means reading the policies as a specialist would; connecting the evidence sources, from repositories and identity systems to contract stores, ticketing, and monitoring; encoding each contract and its mandate with the authorized principal; building adjudicated evaluation sets with the specialists; keeping the producing agent, the verifying agent, and the authorizing mechanism separate; and instrumenting the labor account of Section 10, so that the result is measured in human hours rather than in demonstrations.

The best FDEs will design the automation of the whole DGF, not of a single gate. The difference is not effort, and it is not the order of deployment; it is what gets designed. A coherent end-to-end architecture can be deployed gate by gate: a first gate can run autonomously while using the data model, the identities, and the interfaces designed for the next ones. What must be avoided are automations designed in isolation: local copilots whose outputs are re-extracted and re-verified by the next team, incompatible records, and local savings obtained by moving work elsewhere. Section 11 shows that the total human workload can then rise. An end-to-end design builds one evidence record that every role reads and extends, because the

artifact of one role is the document of the next (Section 5); one case state whose conditions are carried from gate to gate; the route orders the enterprise has chosen, preserved; mandates negotiated once per class of decision rather than once per tool; one assurance architecture; and one labor ledger across the nine populations. The economic content of that difference is the interaction term of Equation (22), which can be positive or negative: the value of designing across gates is the part of the saving that no sum of isolated projects delivers, and it has to be measured, not attributed in advance to the engineer's profile. The second experiment of Section 17 measures it.

Three consequences follow. An FDE hired by one department tends to automate that department's gate and to stop at assistance, because the mandate, the evidence, and the savings that would justify going further belong to other departments; end-to-end automation needs a sponsor above the gates, typically the owner of the DGF itself. The patterns an FDE builds for Buy, Integrate, and Build are reusable across customers, which is the shared-provision mechanism of Section 7.4 in human form: the first deployments are expensive and later ones amortize them. Reuse is one way a low-volume route such as Integrate becomes economical; lowering its exception effort, its review, or its upkeep are others. And the FDE's own work belongs in the account: it is a transition investment while the road is being built and recurring support B_A while it must be maintained. Law 5 applies to it as to any other supervisory work. Report initial implementation hours separately from the recurring FTE account and include them in a cumulative transition-cost assessment. Repeated adaptation, evaluation, and maintenance cannot be relabelled as a one-off investment to exclude them from the 2033 FTE threshold. The title names this implementation perspective; it does not assume that an FDE job title itself predicts a successful deployment.

The profile that matters is therefore not the engineer who can make one agent pass one review. It is the engineer who can read a DGF as a chain of documents and artifacts, redesign the handoffs, obtain the mandates, and prove with the labor account that human work fell across the whole chain. This is the author's assessment of where the scarce skill will lie. It can be checked by comparing, at comparable scope and conditions, the residual ratio ρ of programs designed with common interfaces and one labor account with that of programs designed in silos; deployment can proceed gate by gate in both groups.

20.1 After the DGF: the professions of the business itself

The DGF gathers professions of information technology and its governance that exist in every enterprise, whatever it sells. That is why it comes first in the comparative hypothesis of Section 6, and why it is the natural first road for an FDE: a pattern built for one enterprise's gates is reusable in every other enterprise. The thesis of this section goes one step further. **After the DGF, it will be the turn of the professions that are the business itself: the underwriter and the claims adjuster in insurance, the loan officer and the credit analyst in banking, and so on in each industry.** These professions will be transformed, and many of their posts removed, by the same mechanism, because their work has the same shape: a submission, a claim file, or a loan application is read, and a priced quote, a coverage decision, or a credit decision is returned.

The movement has already begun on the standard path of these professions. In insurance claims, Allianz's Project Nemo coordinates seven specialized agents that handle coverage checks, weather verification, fraud screening, and payout calculation for low-value food-spoilage claims in Australia; the insurer reports processing and settlement times cut by about 80%, from days to hours, with a claims professional still making the final payout decision (Allianz, 2025). That figure is a delay, not a count of human hours; and because a person confirms every payout, the remaining human work sits in the review term of the account rather than in a small

exception queue. In commercial underwriting, AIG has built an agentic system with Palantir and Anthropic: its chief executive described reviewing every submission in some lines without adding underwriters, then a multi-agent design in which agents ingest submissions, evaluate risks against underwriting guidelines, benchmark pricing, and synthesize their findings for an underwriter, who keeps the decision (Insurance Journal, 2025; Reinsurance News, 2026). In consumer lending, Upstart reports that more than 90% of its funded loans are fully automated, with no human involvement required by the company, while its presentation for the same quarter shows about a quarter of evaluated applications going to manual review (Upstart, 2026). The two figures describe different populations with different conversion rates, and the company’s definition of end to end stops at funding for personal loans but at approval for auto loans and home-equity lines, where human action is still required before funding. It is a textbook case of why the accounting boundary of Section 10 matters.

These cases are consistent with the patterns of Section 8.3; they do not measure them. The standard path goes first and a residual remains, but H3 would need the share of baseline labor held by that residual, which none of these sources reports. Analysis is delegated while authorization is retained, the adjuster still releasing the payment and the underwriter still binding the risk, which is compatible with H5 without measuring the two trajectories. None of them establishes H6. And the vendors that deploy these systems work through embedded engineers; the FDE model itself comes from one of AIG’s partners. What is automated today is the simple claim and the small personal loan, not the complex commercial risk, the disputed claim, or the corporate credit. The law predicts that those follow, by the same exhaustion of the residual (H6).

Two differences explain the order. These decisions often concern persons, and the law treats them accordingly: creditworthiness assessment and risk pricing in life and health insurance are listed as high-risk uses in the AI Act (European Parliament and Council, 2024, Annex III). GDPR Article 22 addresses solely automated individual decisions with legal or similarly significant effects, subject to its exceptions and safeguards (European Parliament and Council, 2016). And these professions are vertical: an underwriting pattern is reusable across insurers, not across all enterprises, so the amortization base of Section 7.4 is narrower than for the DGF. Against that, the volumes are far larger than a governance pool of 140 people, so the labor at stake is far greater. That the standard path of these professions is advancing in parallel is also the most direct challenge to the ranking of Section 6, which remains a hypothesis.

The two movements are linked. Every automation of a business profession is itself a project that must pass the enterprise’s gates, as the Build case shows: an automated DGF removes the brake on all the automation that follows it. And the FDE who has closed a whole DGF has learned the method that the business chains require: contracts for each step, one evidence record, mandates by class of decision, an assurance architecture, and a labor account that proves the result. Submission, pricing, binding, and claims in insurance, or application, identity checks, credit decision, and servicing in banking, are roads of the same kind. The same reading applies to a mortgage processor, a customs declarant, or a pharmacovigilance case handler. The labor account of Section 10 is the instrument that will confirm or refute this expectation, profession by profession.

21 Position relative to established work

21.1 The closest economic models

Interdependence is already central to several models of AI automation. Gans and Goldfarb (2026) study *O-Ring Automation*, in which task qualities are complementary rather than separable. Automating one task changes the value of quality in the others. This is an important antecedent:

an automation decision cannot be evaluated only by the task it removes.

Demirer et al. (2026) model ordered steps, manual execution, augmentation, automation, and job boundaries shaped by handoff costs. Their empirical analysis (Section 7) also finds that a step is more likely to be executed by AI when its neighbors in the production sequence are. Combining O*NET tasks, published exposure labels, the Anthropic Economic Index, and model-generated workflow orderings, this is an observational anchor for local propagation, not a causal estimate for enterprise gates. The present paper examines how evidence-bearing handoffs affect downstream human work, retaining verification, exceptions, and authorization in the comparison.

Ide and Talamàs (2025) embed AI in a knowledge economy with hierarchical production and problem solving. The distinction between autonomous AI and assistance to humans changes organizational and distributional outcomes. Our analysis treats a particular governance process as the observation unit rather than solving for labor-market equilibrium. Table 27 locates the difference.

Table 27: Closest models and the empirical object developed here.

Work	Central mechanism	Focus of this paper
Gans and Goldfarb	Complementarity among task qualities.	A labor-accounting framework for execution and assurance at organizational review interfaces.
Demirer et al.	Task chaining, handoff costs, and endogenous job boundaries.	Evidence provenance, exception effort, and authorization within a governed process.
Ide and Talamàs	AI in knowledge-based production hierarchies.	A fixed-process workload and cost comparison before any equilibrium response.

21.2 Established organizational foundations

The gate is not a new managerial construct: Section 4 builds on Stage-Gate and on artifact-centered process models. These approaches explain why an evolving project dossier is a useful unit for tracing work across departments.

Knowledge hierarchies provide the antecedent for the human residual. Garicano (2000) models specialization in solving problems; Garicano and Rossi-Hansberg (2015, Sections 1 and 3) explain how ordinary problems are handled near production while specialists deal with exceptions, subject to the cost of communicating and supplying help. Bainbridge (1983, pp. 775–777) identifies the supervisory, skill-retention, and recovery difficulties created by automation. The selection result of Section 10.2 connects these mechanisms to a measurable share of baseline labor and separates that selection from the effort required after deployment.

Agentic Business Process Management addresses how human and software agents act within process structures. Vu et al. (2026) examine its development and practitioner perspectives. Calvanese et al. (2026) describe process frames, autonomy, explanation, interaction, and adaptation. Our contribution is a measurement framework for the economic consequences of those implementations: a producer-by-format comparison, the complete residual labor boundary, and a quality-constrained forecast of workload.

21.3 What the economics of the firm adds

Coase (1937, pp. 389–394) explains the comparison between coordinating transactions through markets and organizing them within a firm. Review, negotiation, and internal authorization

belong to that coordination problem. Lowering their cost can change the feasible scale and depth of internal control. The direction of the firm’s boundary response depends on external as well as internal coordination costs. Shahidi et al. (2025) analyze the external channel: agents can reduce search, communication, and contracting costs while introducing market frictions. Internal gate savings alone therefore do not determine the firm’s boundary.

Task-based economics remains complementary: substitution can coexist with new tasks and higher demand (Autor, 2015; Acemoglu and Restrepo, 2019). Occupational exposure estimates concern potential task effects (Eloundou et al., 2024); measured AI assistance can have heterogeneous productivity effects even within one occupation (Brynjolfsson et al., 2025). The gate perspective adds the links between these tasks and the work required to trust and authorize their outputs.

These works ground the representation and the mechanisms, but do not establish the universal endpoint proposed here. The distinctive claim is the absence of a permanent human gate class across the gates an enterprise places in its DGF, illustrated here on three generic routes, including authority enactment and the support work introduced by automation. The paper turns that claim into a contract-level thesis, supplies the labor account that measures progress toward it, and specifies a measurement boundary that cannot hide the last human operators.

22 Conclusion: every gate becomes software

The three example routes have different entry points and sequences, and another enterprise would compose its DGF differently, but each route is still composed of transformations of inputs into evidence, decisions, obligations, and authorized actions. The gate-substitution conjecture predicts that all of those gates become machine-executed, including exception handling and support. If that endpoint is attained, the gates, their sequences, and their purposes can remain while the human labor required to execute them contracts. This universal endpoint is an argument of the paper, not a measured outcome.

The case for believing it is cumulative. Gate functions already have information contracts; the DGF combines digital objects, a paved road, decisions about systems, and management pressure as few other expert functions do; several nearby controls already execute as software; agent horizons are increasing; class-level authorization can replace repeated individual decisions; and common provision can fund capabilities that one low-volume enterprise cannot justify alone. None of those facts alone establishes the endpoint. Together they motivate the comparative hypothesis that an expert control function will be completely automated here first.

The path is not a straight line. Identical coverage can leave 43.52 or 168.92 workload-equivalent FTE out of 140, because exceptions, review, rework, and upkeep differ. A residual holding 20% of the cases can hold 40% of the work. The same platform saves on a 95-visit route and loses on a five-visit one. Separating baseline selection κ from post-deployment effort η makes those comparisons coherent, and the account $\phi\eta + (1 - \phi)\mu + r + b_A$ names, route by route, the term that still blocks the endpoint.

The consequence is stated plainly. For a fixed amount of governed work, complete substitution removes the labor requirement that staffed it: from 140 FTE toward a support floor, and, if support closes, toward zero. Redeployment, new projects, and changing firm boundaries determine the uses of displaced capacity; they do not reverse the substitution of the DGF’s execution. Governance can even become stricter while its workforce contracts, because a control whose marginal cost falls can be applied every time.

The dated hypothesis is an 80% reduction in required workload-equivalent FTE in the DGF ecosystem by 20 September 2033 relative to 2026, at comparable governed output, quality, and service. In the illustrative 140-FTE pool, that means at most 28 FTE, including supervision

and support. The existing configuration figures show how the same labor account can lie above or below this threshold; they do not date when any configuration will occur. The test measures human hours, converts them consistently to FTE, and keeps actual staffing and individual headcount as separate outcomes. A ratio above 0.20 rejects the FTE prediction; the current synthetic calculations do not establish its empirical truth.

The companion benchmark provides empirical support for a central technical premise of this argument: models can successfully perform many of the specified governance reviews. Across 899 evaluable model–case runs, strict gate success ranges from 74.18% to 94.98%, while complete-route success ranges from 24.67% to 76.92%. Gemini’s 230 completely conforming routes out of 299 evaluable projects show that its strong performance extends to connected sequences of reviews, under the stated synthetic conditions. These results support the prospect of substantial automation of the represented review tasks. The deterministic control also succeeds on all 1,700 gates and 300 routes, establishing that the supplied rules and facts already suffice to solve this task. The experiment demonstrates model execution capability, without identifying an advantage over that direct rule program in this information condition.

The follow-up clarifies both reliability and failure interpretation. A post-hoc structural evidence criterion raises gate success to 99.06% for Gemini, 85.53% for Luna, and 77.24% for DeepSeek; these are sensitivity results rather than replacements for the strict scores or semantic evidence judgments. Authentically observed CSV quotations can fail the original snapshot-only citation contract. Across 135 repeated runs on 15 existing dossiers, only nine, three, and zero dossiers, respectively, pass all three strict trajectories. Similar aggregate performance therefore does not guarantee stable complete success on every project. Conditional approval and rejected-request counts additionally expose behavior hidden by a final success rate.

The workforce implication is direct but conditional. If comparable performance is achieved in operational settings, automation could replace review work currently performed by employees and reduce the staffing required for a comparable volume of governed projects. This makes workforce contraction a plausible consequence to investigate. Its size depends on the labor remaining for exceptions, oversight, corrections, integration, and maintenance. Correct dispositions alone overstate full review conformity: even the best-performing model fails some evidence checks. No conversion from gate success to an equivalent percentage of FTE savings is justified by these data.

The benchmark supports the technical prospect of task substitution; the 80% FTE reduction by 2033 is the paper’s explicit, testable projection of its potential organizational consequence. Its original 2026 baseline window has closed; testing that exact dated prediction now requires auditable historical records as well as later quality-constrained human-labor observations. No such cohort is enrolled. The FTE prediction can succeed while a smaller human team remains indispensable. Complete execution and support closure remain a longer-term conjecture without a fixed date. After the DGF, the same reading applies to the professions of the business itself, from the underwriter to the loan officer (Section 20.1). The decisive empirical object is not a model’s completion rate. It is the quality-constrained human labor still required by the connected organizational process; the conjecture predicts its direction, while the labor account states what must be measured to assess that prediction.

Data, source material, and code

The workforce and economic scenarios are synthetic. `anc/parameters.json` is the source of their assumptions; `anc/reproduce.py` regenerates their table fragments, figures, and result files, and `anc/test_reproduce.py` checks key synthetic calculations quoted in the text. The ancillary directory also contains mathematical notes, a source map, and the cohort protocol.

Figure 1 is drawn in the manuscript source. Every fact taken from the author’s practitioner framework is reproduced in this paper (Section 9 and Appendix A). The METR series is cited, not re-estimated; the three-hour 80% anchor is a declared scenario input. No confidential enterprise logs or autonomous deployment results are used. Supplied figures and table fragments permit L^AT_EX compilation without Python. Ancillary paths above refer to the public `paper/anc/` directory; these reproducibility materials are available in the repository rather than required for the minimal L^AT_EX upload.

The measured model outcomes in Section 18 have a separate data source. The public GitHub release `dgf-bench-300-20260923` contains the complete recorded experiment directory, its 300-case dataset (including evaluator-only reference files for offline reproduction), and a snapshot of the benchmark sources used by the run. Archives preserve file bytes, including historical failed attempts and superseded reports; per-file inventories and archive SHA-256 checksums support verification. During evaluation, the agent tool interface excluded the reference files. Their subsequent publication does not imply they were agent inputs. The repository directory `research/2026-09-dgf-bench/` contains the final tables, diagnostics, paired comparisons, figure, and offline reproduction script. Reproduction from recorded scores makes no model calls. Rerunning paid inference is a different operation and need not produce identical outcomes.

The same release now contains `dgf-bench-repetitions-20260924.zip`, preserving all 4,020 files of the completed 135-run follow-up, and the evidence-audit and review-follow-up archives. These include original excerpts, observed sources, conditional-approval records, the 300-dossier source inventory, counterexample documents, and three development dossiers. The original 300-project run, dataset, source snapshot, and primary scores are unchanged. The API expenditure is USD 87.015829172 for the original experiment and USD 12.5598870228 for the repetitions, totaling USD 99.5757161948. Offline controls and audits make no new model calls. Prepared General-replay inputs and source-location prototypes are explicitly separated from executed model results; the new replay and policy/evidence ablations have not run.

The focused companion manuscript, *DGF-Bench: Rule Application and Evidence Reliability in Synthetic Governance Reviews*, reports the same original and repeated model data, with extended evidence diagnostics and model-evaluation methods. Shared methods and results are not independent evidence; this paper develops their separate relationship to gate contracts, FDE implementation, and labor-accounting hypotheses.

Authorship and use of AI tools

Jeremy Canale is the author and is responsible for the research claims, source checks, and interpretation. Generative AI assistance was used for drafting and revising text, developing supporting code, and examining recorded experiment outputs. The tested model endpoints and their inference settings are identified in Section 18. Automated assistance and deterministic consistency checks do not constitute independent expert adjudication.

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A DGF source correspondence

The source is Canale (2026), a 124-page unpublished practitioner framework. The paper uses it only for the facts reproduced in Section 9 and located in Table 28: the three initial situations, the opinions recorded on them, the bounded acquisition exception, the route types, and the ordinal risk scale. Nothing else in the framework is needed to follow or to check the argument. Page numbers refer to the printed source. The gate types and example tasks of Table 1, the life-cycle positioning of Table 2, and the illustrative reviews of Table 5 summarize the author’s governance practice; they are not taken from the pages listed below. The nine populations, the 140-person allocation, route volumes, labor-displacement fractions, support floors, calendar scenarios, and proposed machine operations are this paper’s analytical additions. They are not measured staffing or performance data from the framework.

Table 28: Framework locations used by the argument and the three worked routes.

Printed pages	Material used
14, 16 17–20	Dispositions and responsible functions referred to in the three cases. The three route types (new platform, acquisition, new project) abstracted here as Buy, Integrate, and Build.
21–23, 28, 32–36	Required evidence, dated obligations, procurement exchanges, shared IT/Security dossier.
38, 90, 98, 100–101	Business risk ownership, operational handover, ordinal risk scale, exception process.
103–105	CRM red opinion, bounded acquisition exception, support-agent red opinion.

Shared work, cross-gate negotiation, and renewed review require additional visit records in an empirical implementation.

B Readiness, interfaces, and process accounting

B.1 Prospective readiness measurement

For a case class, enumerate required facts, decision criteria, evidence claims, actions, and acceptance tests independently of the evaluated agent. Define $d_i = (O_i, C_i, V_i, X_i, Y_i) \in [0, 1]^5$ using Table 29. Undefined denominators are reported as such. They do not become perfect scores.

Table 29: Candidate readiness dimensions. Ratios are measured before evaluated outcomes.

Score	Numerator / denominator	Distinction
O	Accessible required facts / all required facts.	Authorized accessibility, not truth.
C	Explicit testable criteria / enumerated criteria.	Auditable requirements and ambiguity handling, not a complete encoding of expert thought.
V	Claims with independent checking routes / material claims.	Verification access, not a claim that every check passed.
X	Executable scoped actions / required actions.	Technical capability, not authority to approve.
Y	Outcomes with independent tests or adjudication / required outcomes.	Evaluation coverage, not observed success.

A candidate summary is $A_i = (O_i C_i V_i X_i Y_i)^{1/5}$. It is a descriptive readiness index, not a success probability. Its value for $(0.01, 1, 1, 1, 1)$ is about 0.398; a minimum index and individual hard blockers provide useful alternative predictors. An unavailable mandatory action can block deployment even if most scored actions are accessible. Institutional permission Γ_i is recorded separately. The formalization need not encode every step of tacit expert reasoning: independent outcome adjudication can exist where procedural rules are incomplete.

For an artifact change $\Delta z_{ij} = \max(0, z_{ij}^1 - z_{ij}^0)$, a bounded positive-change representation is

$$d'_{ja} = d_{ja} + (1 - d_{ja})\beta_{ija}\Delta z_{ij}, \quad 0 \leq \beta_{ija} \leq 1. \quad (32)$$

Here z_{ij} is normalized to $[0, 1]$, so that $\Delta z_{ij} \in [0, 1]$ and the updated dimension stays in $[0, 1]$. Only dimensions affected by the artifact receive nonzero sensitivity. The readiness index is recomputed from the new vector. An unchanged artifact makes no change, an unaffected zero dimension remains zero, and authority remains separate. Degradation is evaluated as its own outcome rather than inferred from this positive-change representation. The labor response remains Equation (16) and requires estimation.

B.2 Visits, allocation, and baseline coherence

For a stable expected-flow model, $\lambda = v + R^\top \lambda$, hence $\lambda = (I - R^\top)^{-1}v$ when the spectral radius of R is below one. v counts exogenous visits and R_{ij} expected follow-on visits to j from one visit to i . Branching and remediation can make visits exceed new projects. Direct logs are preferred when an expected routing model is inappropriate.

The synthetic case partitions in Table 30 satisfy Equation (8). The frozen partition separates baseline concentration from subsequent inflation; post-deployment multipliers are $k = \kappa\eta$. The reproduction code checks this constraint, the four workload totals, the fragility margins, and the route illustration.

Table 30: Baseline partition, symmetric review effort, and fragility margin. $q\kappa = \phi$ is retained baseline case labor. $\mu = h_R/h_S^H = m/\sigma$ measures review against its own group’s baseline, with $m = 0.10$ in S0–S2 and $m = 0.35$ in S3. η^* is the exception-effort ratio at which workload equals the baseline under the S2 review, rework, and upkeep (Equation 13).

Pool	q	κ	σ	$q\kappa$	μ (S0–S2)	μ (S3)	η^* (S2)
R1	0.22	2.00	0.7179	0.44	0.1393	0.4875	1.77
R2	0.33	2.00	0.5075	0.66	0.1971	0.6897	1.20
R3	0.18	2.00	0.7805	0.36	0.1281	0.4484	2.42
R4	0.16	2.00	0.8095	0.32	0.1235	0.4324	2.64
R5	0.22	2.00	0.7179	0.44	0.1393	0.4875	1.77
R6	0.24	2.00	0.6842	0.48	0.1462	0.5115	1.81
R7	0.26	2.00	0.6486	0.52	0.1542	0.5396	1.60
R8	0.33	2.00	0.5075	0.66	0.1971	0.6897	1.20
R9	0.33	2.00	0.5075	0.66	0.1971	0.6897	1.20

C Demand, composition, and an information limit

C.1 Demand and composition do not disappear

At common scaled demand $g(t)$, variable labor is proportional to $g(t)h(t)$. Section 12 gives the consequence: a demand response is not a failure of gate-level substitution. New governing requirements can add contracts, while recurring requirements necessary for an existing contract remain in its support account. The fixed 2033 workforce deadline prevents new terminology from resetting the dated prediction; new DGF roles remain in its accounting boundary. A changing economy prevents an old task registry from proving a claim about all future jobs.

C.2 Why writing inputs and outputs is not an existence proof

Suppose two possible environments produce the same accessible input history but require disjoint correct outputs. Any deterministic system restricted to that history returns the same output in both and must fail in at least one. A randomized system cannot guarantee success in both either. Additional inquiry or a contract that permits an explicit unresolved disposition is necessary. A human with access to new information changes the information boundary, not this logical constraint.

This example identifies a concrete test of the law: can the missing observation or accepted decision procedure be supplied without indispensable human execution? The paper predicts eventual coverage of the DGF functions; it does not infer coverage from input-output notation alone.

D Illustrative substitution configurations

This appendix shows what the labor account returns for chosen substitution fractions. It is an arithmetic illustration of Section 12, not an estimated trajectory or a forecast calendar. Take the 140 people of Table 3 with $K = 120$ hours per month and full allocation across the three routes. Table 31 gives an assumed split of each population’s baseline FTE between W1, W2, and W3. The modeled pool is the buying and integrating enterprise’s governance workforce; supplier-side production in Buy requires a separately measured supplier account.

Let a_w be the fraction of baseline case-processing labor displaced on route w , after counting remaining case review and rework, and F additional shared human support in FTE. Then

$$N_A = 51.25(1 - a_1) + 36(1 - a_2) + 52.75(1 - a_3) + F. \quad (33)$$

Table 31: Synthetic populations shared across the three workflows. The route columns are allocated baseline FTE, not additional employees.

Pool	Population	Baseline	W1	W2	W3
R1	Procurement	10	7.50	1.00	1.50
R2	Legal / Contracts	10	5.00	3.00	2.00
R3	Architecture	30	9.00	7.50	13.50
R4	Security Architecture	20	6.00	5.00	9.00
R5	Compliance / Privacy / GRC	10	4.50	2.50	3.00
R6	IT Production and Standards	25	6.25	7.50	11.25
R7	Tech Readiness	15	6.00	3.00	6.00
R8	General Gate Support (PMO, Portfolio, Risk)	10	3.00	4.00	3.00
R9	Finance / Cost Control	10	4.00	2.50	3.50
Total		140	51.25	36.00	52.75

The uneven-substitution row of Table 32 leaves 30%, 75%, and 25% of baseline route labor, reflecting the route order proposed in Section 15; the advanced row leaves 5%, 20%, and 3%. Each uses five shared support FTE. The final rows distinguish completion of direct gate execution from closure of its remaining support work. Totals use unrounded values and are displayed rounded half up.

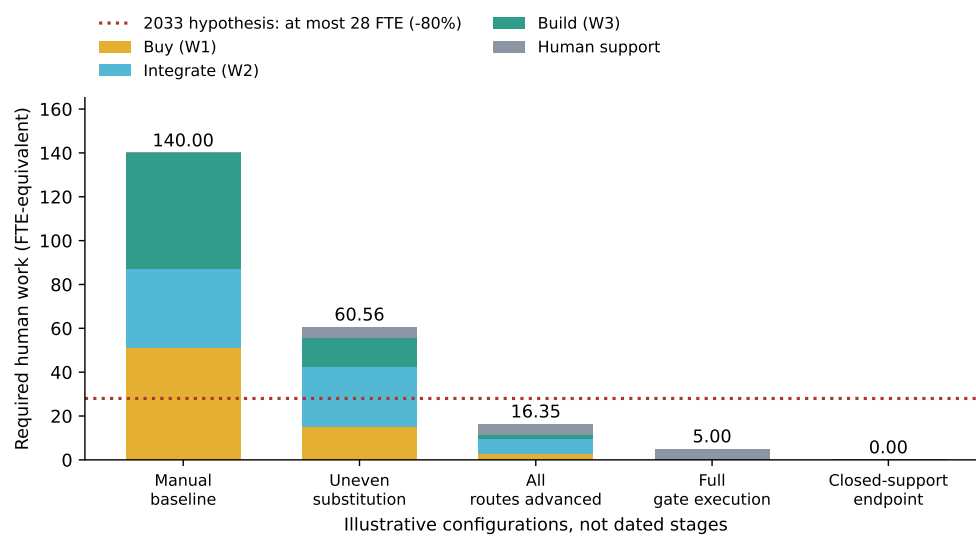
Table 32: Required human workload (FTE-equivalent) in illustrative substitution configurations. All three workflows remain in the process.

Operating configuration	W1	W2	W3	Support	Total
Manual baseline	51.25	36.00	52.75	0.00	140.00
Uneven substitution	15.38	27.00	13.19	5.00	60.56
All routes advanced	2.56	7.20	1.58	5.00	16.35
Full gate execution	0.00	0.00	0.00	5.00	5.00
Closed-support endpoint	0.00	0.00	0.00	0.00	0.00

With common labor displacement a across routes, Equation (33) becomes $N_A = 140(1 - a) + F$. Table 33 shows how a support floor determines the endpoint.

Table 33: Residual required FTE under common labor displacement and alternative human-support floors. These are conditional arithmetic values.

Labor displaced	No support floor	5 FTE floor	15 FTE floor
0%	140.00	145.00	155.00
50%	70.00	75.00	85.00
90%	14.00	19.00	29.00
99%	1.40	6.40	16.40
100%	0.00	5.00	15.00



Synthetic calculations | Support included | The 2033 threshold is a hypothesis, not an observed result

Figure 8: The workflows remain; their execution workforce contracts. Route contributions are shown separately and the shared human-support term is counted once. The final zero is the consequence of complete closure assumed in that configuration, not a measured capability or a predicted date. The dotted 28-FTE line marks the separate 2033 hypothesis: 80% below the illustrative 140-FTE baseline, support included. The bars are configurations, not a calendar trajectory.

E A worked gate-contract specification (illustrative)

Seventeen boxes in a figure are not seventeen benchmark contracts. Table 34 works one contract through, the Tech readiness gate of the Build route before production, as a model for the others. It is an illustration written for this paper, not a contract from the framework; its thresholds are examples. The same gate also intervenes at framing, to assess the maturity of candidate technologies (Table 2); that intervention is a separate contract version with its own inputs and criteria. Closing this contract would close one case population and one intervention of the Tech readiness gate, not the gate class.

Table 34: A worked contract specification: Tech readiness on the Build route, before production.

Field	Specification
Identity	Route W3 (Build), gate Tech readiness, pre-production intervention; contract version and policy version recorded on every visit.
Admissible cases	Internal builds hosted on enterprise platforms, classified up to confidential. Safety-critical systems are outside the declared population: they are refused with a reason, and a refusal of an admissible case counts as a false refusal.
Accessible information	Revised design after security re-review; reservations register of earlier gates; pilot results; load-test report; code-quality report; recovery, resilience, and backup dossier (backup policy, last restoration test report, recovery objectives); runbooks; monitoring plan. Read access to the configuration database, delivery pipeline, and ticketing.
Tools and actions	Read repositories, pipelines, and test reports; launch a restoration test in the pre-production environment; request evidence from the project; open and close reservations. The agent cannot modify tests or thresholds, change policy, or approve its own exceptions.
Decision criteria	Load test at or above the target volume with latency under its threshold; restoration test less than twelve months old with measured recovery time within its objective; code-quality gate passed; every blocking reservation of earlier gates lifted with evidence; runbooks present and exercised. Thresholds are read, not embedded: target volume and latency from the versioned non-functional requirements of the project, recovery objectives from the service record in the configuration database. The version referenced by the reservations register at the date of the visit is authoritative, and each value is written to the trace with its source.
Allowed dispositions	Ready; ready with dated reservations; not ready, with the rework required; evidence requested, leaving the case unresolved; refusal as out of scope.
Quality margins	Against an independently adjudicated reference, with stated denominators: unsupported “ready” among visits adjudicated not ready, ceiling 10^{-3} ; false “not ready” among visits adjudicated ready, ceiling 5%; omitted blocking findings among the blocking findings of the reference, ceiling 1%; no leakage of dossier content; 95% of visits decided within two working days. Acceptance requires a one-sided 95% upper confidence bound below each ceiling on a qualification set whose size is fixed in advance, for instance 2,995 failure-free not-ready cases for the first ceiling.
Mandate	Standing delegation from the production release committee for the admissible case class; principal: the chief technology officer; validity twelve months; suspended by any unsupported “ready” or by a drift alarm; cases outside the mandate stay unresolved or are refused. Renewal, and reinstatement after a suspension, are decided by the same committee on a requalification report produced from the evaluation set; that committee time and any human requalification work are recorded in B_A .
Human work counted	Per visit: exception handling h_E , sampled review h_R , rework h_W . Recurring support B_A : maintenance of criteria and thresholds, upkeep of the evaluation set, incident review. Supplier labor, for instance from a test-tooling vendor, is recorded in a separate supplier account. Shared platform support is allocated once, by a stated rule such as visits, and never counted in two contracts.
Qualification plan	Volume from historical dossiers replayed off production, dedicated campaigns, and several sites; a statement of why those cases are representative and independent; drift monitoring after deployment; requalification after any change of policy, tools, or model.

F Notation and units

Symbol	Meaning and units
W1, W2, W3	Buy, Integrate, and Build route schemas (illustrative).
$G_{w,j}, C_{w,j}$	Route-position work package and its input-output/action contract.
$S, E, \Pi, w; D, U$	Case state, accessible evidence, versioned policies, route context; disposition and action/evidence trace.
P_d, α_{di}, K_d	People; allocation fraction; useful hours/person/period.
λ_s, L_s, N_s	Visits/period, human-hours/period, and workload-equivalent FTE.
h_0, h_E^H, h_S^H	Overall and conditional baseline human-hours/visit.
h_E, h_R, h_W, B_s	Post-deployment exception, review, extra rework hours/visit; fixed human upkeep hours/period.
$q, J, e(T); T, \phi$	Exception share, residual indicator, routing propensity; baseline case hours and the fraction of baseline case labor in the residual.
κ, σ, η, k	Baseline conditional effort ratios; exception effort multiplier; $k = \kappa\eta$.
μ, m, r, b_s, ρ	Standard-group review ratio; $m = \sigma\mu$; rework, upkeep, and total residual/baseline workload ratios.
q^*, k^*, η^*	Break-even exception share, exception effort, and exception-effort inflation.
$F_s, F_c, N; B_s, B_c$	Shareable and customer-specific fixed cost, number of customers; shared and local upkeep hours (Section 7.4).
$\nu_i, \ell_i, B(t); n^*, u$	Visits, human-hours per visit, recurring support over time; cost-minimizing headcount and target utilization.
a_w, F	Fraction of baseline route labor displaced; additional shared support in FTE.
M, ϵ	Maximum compliant route visits and conditional per-visit failure bound.
$H_{80,0}, d, D$	Anchor 80% horizon, doubling interval in months, target task length.
e_0, e^*, z, ν, c	Initial and target error, tenfold reductions, months and cost factor per tenfold reduction (Section 15).
z, x, θ, Π	In the interface model: artifact characteristic, case mix, model/tool configuration, routing and assurance policy.
$d_i, A_i, \Gamma_i, \beta_{ija}$	Readiness vector/index, authority indicator, and interface sensitivity.
w_s, a_s, F_s, ℓ_s	Monetary units/hour, nonlabor units/visit, fixed nonlabor units/period, attributed expected error loss/visit.
$\bar{c}_s, T_s, J_s, v_s$	Average cost/visit, total period cost, fixed cost, and variable cost/visit.
$C_s, \mathcal{L}_s, B(S); V_t, I_t, d$	Pipeline cost, aggregate incident loss, intervention-set saving; benefits, expenditure, discount rate.
$\xi, u, g, \varepsilon, \tau, a$	Strategic model: true compliance, genuine remediation, cosmetic reporting, score noise, threshold, assurance intensity.
$V, c_u, c_g, D; p, f$	Approval value, effort costs, expected detection consequences; approval probability and noise density. $\omega_u, \omega_g, t, \Delta$ are local curvature terms.
$\mathcal{C}_i$	Correctness event for gate i .
$A_{cj}, F_{cj}, Y_{cj}^{(v)}, \beta^{(v)}$	Producer, format, outcome, and coefficients for hours ($v = h$) or cost ($v = c$).

Some letters are reused across sections with the local meaning given there (z, ν, d, F, a). For the dated FTE hypothesis, $\bar{N}_y$ denotes mean monthly required FTE over the year ending on 20 September of year y , at comparable governed output. All scenario labor values use a common $K = 120$ for transparency.